

CHAPTER ONE

1. Introduction

Pathology is the precise study and diagnosis of disease. The word pathology is from Ancient Greek **pathos** “feeling, suffering” **logos** “the study of”

Plant pathology /phytopathology- It is defined as the study of the organisms and environmental conditions that cause disease in plants, the mechanisms by which this occurs, the interactions between these causal agents and the plant (effects on plant growth, yield and quality), and the methods of managing or controlling plant disease.

The objectives of plant pathology are to study:

- The living entities that cause diseases in plants;
- The non-living entities and environmental conditions that cause disorders in plants;
- The mechanisms by which the disease causing agents produce diseases;
- The interaction between the disease causing agents and the host plant in relation to overall environment; and
- The method of preventing or management the diseases and reducing the losses/damages caused by diseases.

Role of plant pathology

- Preserve the environment
- Secure global food security more importantly in developing country
- Help grow healthy crops
- Maintain the beauty of ornamental plants
- Invent new ways to control/manage plant diseases
- Have rewarding professional careers

History of Plant Pathology

Globally, enormous losses of the crops are caused by the plant diseases. The loss can occur from the time of seed sowing in the field to harvesting and storage. For example important historical evidences of plant disease epidemics include:-

- Irish famine due to late blight of potato (Irish famine, 1845)-1.5 million Irish died
- Bengal famine due to brown spot of rice (Ireland, 1942), and
- Coffee rust (Sir lanka, 1967).
- Two forest tree diseases Dutch elm disease and chestnut blight large economic loss in America. Such and similar epidemics had left their effect on the economy of the affected countries.

Significance of Plant Disease

- kinds and amount of losses
- may limit the kinds of plants and industries in an area
- reduce the quality and quantity of plant produce
- may make plants poisonous to humans and animals
- cause financial losses

Development of plant pathology as a science:-passed through different phases

- **Spiritual phase-** believing that disease is a result of sins-more sins more disease-disease e.g. The ancient Romans sacrificed red animals such as dogs, foxes, and cows to the rust god (tomb with animal picture), Robigo or Robigus, each spring during the festival called the Robigalia in hopes that the wheat crop would be spared from the ravages of the rust
- **Descriptive phase-**familiarity with existence of a disease- description
- **The experimental phase-** design experiments to answer questions like how pathogens enter host, multiply, and spread within the plant, the mechanism of host plant cell death and breakdown, pathogen sporulations, spore dispersal, overwintering and over summering, germination, effect of environment and disease development etc
- **The etiological phase-**involved observations and experiments aimed at proving the causes (etiology) of specific plant disease-Koch's postulate
- **Search for remedy phase-** attempting to find management options for the disease

- **All rounded research style-** basic research, applied, creative and strategic research to increase understanding of plant diseases

Concepts of plant disease

What is Disease?

- A malfunctioning process that is caused by continuous irritation. Of course, this process must result in some suffering, and produce symptoms
- The term plant disease is properly applied to any deviation from normal growth or structure of plants that is sufficiently pronounced and permanent to produce visible symptoms or to impair quality and economic value.
- Any disturbance of a plant that interferes with its normal growth and development, economic value, or aesthetic quality; a continuously, often progressively affected condition in contrast to injury, which results from momentary damage.
- Any disturbance brought about by a pathogen or a consistent environmental factor which interferes with manufacture, translocation, or utilization of nutrients
- Failure to reach full genetic potential due to the activities of another organism or environmental factor.

Scope of Plant pathology

Plant pathology comprises with the basic knowledge and technologies of Botany, Plant anatomy, Plant physiology, bacteriology, Virology, Nematology, genetics, molecular biology, genetic Engineering, Biochemistry, Horticulture, Tissue culture, Soil science, Forestry, Physics, Chemistry, metrology, Statistics and many other branches of applied science.

Causes of Plant Diseases

Plant diseases caused by pathogens. Hence a pathogen is always associated with a disease. In other way, disease is a symptom caused by the invasion of a pathogen that is able to survive, perpetuate and spread. Further, the word ‘pathogen’ can be broadly defined as any agent or factor that incites ‘pathos or disease in an organism or host. In strict sense, the causes of plant diseases are grouped under the following categories.

1. **Animate (biotic causes):** - Pathogens of living nature are categorized in to the following groups.
 - Fungi
 - Bacteria
 - Protozoa
 - Nematodes
 - Parasitic higher plants
2. **Mesobiotic causes:** - The disease incitants are either living or non-living
 - Viruses
 - Viroides
3. **Inanimate (abiotic cause):** - In true sense these factors cause damages (any reduction in the quality or quantity of yield or loss of revenue) to the plants rather than causing disease. The causes are:
 - Temperature
 - Moisture
 - Light
 - Deficiencies or excess of nutrients
 - Lack of oxygen (e.g. Hollow and black heart of potato)
 - Toxicity of pesticides, herbicide burning
 - Air pollutants (e.g. Black tip of mango)
 - Improper cultural practices –mechanical damage
 - Abnormality in soil conditions (acidity, alkalinity)

Chapter Two

Classification of Plant Diseases

Why we need to classify plant diseases?

To facilitate the study of plant diseases they are needed to be grouped in some orderly fashion. Various schemes of classifying plant diseases have been presented in plant pathology in a logical system, depending on the purpose of treatment. The following are the bases of plant disease classifications:

- a) Host plants or crops affected (crop category or taxonomic approach) -e.g. cereal diseases, pulses, oil seed, vegetable, ornamental, forest disease, etc.
- b) Affected parts (organs) of the plant (Histological approach) - e.g. aerial disease, root, stem, leaf, floral, fruit, seed, etc, diseases.
- c) Based on symptoms (signs) (symptomatology approach) - e.g. rots, rust, mildew, blights, smut, etc.
- d) Based on nature of infection (epidemiological approach)
 - **Endemic:** The diseases are more or less constantly present from year to year in a moderate to severe form.
 - **Epidemic/epiphytotic:** Diseases that occur widely but periodically. E.g. late blight of potato.
 - **Sporadic:** Diseases that occur at very irregular intervals and location, causes minor loss. e.g. Stem rust (*Puccinia graminis*)
 - **Pandemic:** Diseases that result in wide devastation and starvation (famine) leading to death of human beings. E.g. late blight of potato (*Phytophthora infestans*) in Ireland in 1845.
- e) Based on source of inoculum/spread mode: E.g. Air-borne (wind-borne), soil-borne, water-borne; Stem rust of wheat is an air-borne disease.

But the most **useful** classification has been made based on the type of pathogens that cause plant diseases. Since this type of classification indicates not only the cause of the disease, but also the knowledge and information that suggests the probable development and spread of disease along with their possible control measures.

The classification is as follows:

1. Infectious plant diseases

- A. Disease caused by parasitic organisms: biotic factors
- B. Diseases caused by viruses and viroids.

2. Non-infectious or non-parasitic or physiological diseases: abiotic factors.

2.1. Infectious causes of plant Diseases (Etiology)

2.1.1. Fungi as pathogens

Fungi: are microscopic, eukaryotic, usually filamentous, spore-bearing organisms; **fungi** (Singular: **fungus**) are classified within their own kingdom. A fungus is neither a plant nor an animal. It is similar to a plant, but it has no chlorophyll and cannot make its own food like a plant can through photosynthesis. They get their food by absorbing nutrients from their surroundings.

Structure/morphology

- The thread like filamentous of the fungus is known as hyphae (sing. Hypha).
- Hyphae of one body or thallus are known as a mycelium (pl. Mycelia).
- Spores are the reproductive or propaetive bodies of fungi. In some fungi, spores of the anamorph (asexual) stage are formed by cleavage of cytoplasm with in a sack or sporangium (pl. sporangia). Spores produced in the sporangium are referred to as sporangiospores.

There are about 100,000 fungal species in the world. More than 8,000 fungal species cause diseases in plants, each attacking one or many kinds of plants. Thus fungi can cause about 100,000 plant diseases. It could be obligate (bitrophs) or non-obligate parasites.

✓There are two modes of nutrition in fungi:

1. **Saprophytes:** organisms that live on dead bodies; they can produce enzymes, cellulose, to decompose the bodies of plants and changed to organic matter.
2. **Parasites:** live on the body of living organisms known as hosts. Two types of parasites:-
 - a) **Endoparasites:** organisms found inside the body of the plant. E.g. Downy mildew on onion
 - b) **Ectoparasites:** outside the body of the plant. E.g. Erysiphae gramineae → caused by powdery mildew on cereals.

Habitat of Fungi (Ecology)

Fungi exist in various habitats, including deep down in the ocean, lakes, rocks, deserts, very salty environments, and areas of extremely high or low temperatures.

E.g. Powdery mildew fungal spores can germinate at very dry conditions ⇨ can survive under varied situations.

★ Generally, most plant pathogenic fungi usually favored by **cool** and **humid** conditions.

Reproduction of Fungi

Fungi chiefly reproduce by means of spores (sporulation). The process is known as cryptogamy.

A. Spores: are specialized reproductive (propagative) organs of one or a few cells. Ex. Rust pustules- spore of rust fungi.

Spores can be used as - reproductive organs, Survival, Dispersal, Infection.

B. Fruiting bodies: usually contains certain organs or produce structures involved in reproduction. Sporangiophores: extension of mycelium

Conidiophores: conidia is the spore

There are two types of spores

1. Asexual spores

Asexual spores may be produced directly from the hypha in some fungi - without the need for fruiting bodies. The spores then germinate and produce additional mycelium, which spreads rapidly. Experts say this allows more rapid dispersal than sexual reproduction.

A) Zoospores: Motile spores that can swim by means of flagella (endogenous).

B) Conidia: Produced by the cutting-off of terminal or lateral cells from special hyphae called conidiophores (exogenous spores). The process is known as conidiogenesis and conidia are released from the conidiogenous cells, e.g. Phialides. Conidia vary in **shape, size, color** and **number of cells**.

C) Chlamydospores (resting structures): Modified hyphal cells with a thick wall (exogenous). These could be apical or intercalary chlamydospores.

D) Pycniospores/Pycnidiospores: endogenous spores produced in the pycnia/pycnidia.

◀Pycnidia is Flask shaped and contains conidia and conidiophores,

◀Pycnidia usually have a hole (ostiole) from which conidia are pushed out of the structure.

2. Sexual Spores

✓ Sexual reproduction occurs in most groups of fungi.

✓ Fungi are usually classified by their teleomorph (sexual) stage.

A. Zygosporangia: gametes (sex cells) unite to produce a zygote- Exogenous cell. The fusion of gametes that are of **equal size** and **appearance** results in **a zygote** referred to as a zygote and are classified in the zygomycota.

B. Oospores: Product of fusion of **unequal sex cells** (dissimilar gametes) –exogenous

C. Ascospores: Usually **eight in number**, are produced within a zygote cell, the ascus (pl. asci) - sexual and endogenous spores produced in ascus. Asci may be produced naked (not in any structure) or in specialized structures.

- D. **Basidiospores:** produced on the outside of the zygote cell or basidium (pl. basidia); **usually four in number.** E.g. rusts and smuts-produced (born) naked or formed in structures like mushrooms, puffballs and conks- exogenous spores.

When both female and male gametes are produced on the same mycelium called **hermaphroditic fungi**. When the male gametes can fertilize the female ones of the same mycelium called **Homothallic fungi (Isogamy)**. When the male gametes fertilize only the female gametes of another, sexually compatible mycelium called **Heterothallic fungi (heterogamy)**.

Classification of Plant Pathogenic Fungi

The fungi that cause disease on plants are a diverse group. Organisms that were collectively grouped as fungi have been placed in the following three different kingdoms:

Lower Fungi

1. Stramenopila(Chromista)

- Phylum-Oomycota → class Oomycetes.

2. Protozoa

- Phylum-Plasmodiophoromycota → class-Plasmodiophoromycetes.
- Myxomycota → class- mycomycetes.

3. Fungi:

- Phylum –Chytridiomycota
 - Zygomycota
 - Ascomycota
 - Basidiomycota

Fungi like organisms or Pseudo fungi (The lower Fungi)-The classification of organisms (taxonomy) including fungi are based on criteria that changes as more information becomes known about them. The taxonomy of fungi has been based on:

- ✓ Sexual,
- ✓ Asexual spore morphology and How they are formed followed by Secondary considerations- such as hyphal and colony characteristics.

- ✓ Molecular techniques that examine and compare the genetics of organisms are increasing in importance to taxonomists.

As previously mentioned, the “fungi” as a group used to be classified with the plants

In the kingdom Planta, but are now placed in three different kingdoms.

1. Kingdom Protozoa: The endoparasitic slime molds belong to this group. Microorganisms may be unicellular, colonial, very simple cells

Phylum -Mycomycota- produces plasmodium like structures.

Class- Myxomycetes (slime molds) body is naked, amorphous plasmodium, produce zoospores, and do not infect plants.

Order – Physarals

Genus – Fuligo, Mucilago and Physarum.

Phylum – Plasmodiophoromycota

Class - Plasmodiophoromycetes

Order- Plasmodiophorals

Genus- Plasmodiophora. P. brassicae causing club root of crucifers.

2. Kingdom Straminopila (Chromista): The water mold fungi (also called straminopiles) belong to **Phylum - Oomycota**

- ✓ Unicellular or multicellular
- ✓ Filamentous or colonial
- ✓ Contains algae, diatoms, oomycetes and some other similar organisms
- ✓ Possess cell wall

Phylum Oomycota: have flagellate zoospores

Class: Oomycetes (water molds, white rusts and downy mildews), Have non-septate elongate mycelium.

The True Fungi/ Higher Fungi

3. Kingdom Fungi: Produce mycelium, that walls of which contains glucans and chitin, lack chloroplast.

Phylum Chytridiomycota: - produce zoospores that have a single posterior flagellum.

- ✓ The chytrids inhabit water or soil.
- ✓ The mature vegetative body transfers in to thick walled resting spores or a sporangium.

- ✓ They are coenocytic and lack a true mycelium.
- ✓ The thallus is irregular, globose, or ovoid; diploid.

Class –Chytridiomycetes

Genus – *Olpidium*; *O. brassicae*: - an endobiotic parasite of the roots of cabbage and other monocot and dicots.

Olpidium is a known vector of plant viruses such as Tobacco necrosis virus and lettuce big vein virus.

Phylum Zygomycota

- ✓ Well-developed, coenocytic, haploid mycelium with chitin in their cell wall.
- ✓ Produce non-motile spores called sporangiospores (produce in sporangia).
- ✓ No zoospores, the resting spore is a zygospore, produced by fusion of two morphologically similar gametangia of opposite mating type (heterothallic).
- ✓ Weak pathogens, terrestrial, and saprophytes causing soft rot or molds.

Class – Zygomycetes (bread mold) – Saprophytic or parasitic of plants, animals and humans. E.g.

Rhizopus stolonifera (bread mold).

Order – Mucorales: Non-motile asexual spores.

Genus- *Rhizopus*: bread mold and soft rot of fruits and vegetables.

◀ **Phylum Ascomycota**

- ✓ With well-developed, septate mycelium containing chitin in their cell walls.
- ✓ The sexual structures of the ascomycetes is an ascus, which is shaped like a sack and contains the products of meiosis, called ascospores, which is normally number eight per ascus.

Ascospores are forcibly liberated from asci, which are produced in fruiting structures called ascocarps. Ascocarps produced by various ascomycetes include the following:

- **Cleistothecia:** Completely closed ascocarps lined with one or more asci. E.g. obligate parasites (*Erysiphe*- powdery mildew).
- **Perithecia:** Flask-shaped ascocarps with an ostiole at the tip.
- **Apothecia:** Open, cup-shaped ascocarps where asci are exposed.
- **Ascostroma or Pseudothecia:** asci produced in a cavity or locule buried within a matrix (stroma) of fungal mycelium.

-Asci not produced in an ascocarp are called naked asci. E.g. Yeast fungi.

Phylum Deutromycota (Deutromycetes, Imperfect and Asexual Fungi)

- ✓ Not true phylum (artificial group)
- ✓ Have no known sexual state in their life cycle; mitotically/via conidia.
- ✓ They have hyphae that possess the characteristics of the teleomorph.

Phylum Basidiomycota

- ✓ They are true fungi that often have large, macroscopic fruiting bodies.
- ✓ They have well-developed, septate mycelium with chitin in their cell wall.
- ✓ The mycelium of basidiomycetes can be dikaryotic, each cell containing two haploid nuclei.
- ✓ The sexual structure is called a basidium or (club).

Class- Basidiomycetes (club and mushroom fungi)

Basidiospores are produced externally on a club like, one or four celled spore producing structure called a basidium.

Order – Ustilaginales (the smut fungi)-Basidium has cross walls or is non-septate.

Genus – *Ustilago* –causing smut of corn (*U. maydis*), loose smut of oats (*U. avenae*), of barley (*U. nuda*), and wheat (*U. tritici*).

2.1.2. Bacteria as a pathogens

Bacteria are single-celled, prokaryotic microorganisms, lacking chlorophyll and characterised by rapid reproduction. Typical bacterial cells are about 1µm in width compared to a eukaryotic cell and are everywhere in nature and physiologically diverse and they occupy a wide range of ecological niches. Depending on the environmental factors like moisture, temperature, oxygen level, they multiply rapidly in nature.

There are about 1600 bacteria species in the world. The majority are saprophyte. About 180 species of bacteria causes plant diseases. Most phytopathogenic bacteria are facultative saprophytes and grow on synthetic media.

Morphology and Physiology of Bacteria

Bacteria differ in morphology and physiology: size, shape, motility, color, Gram stain, etc.

Shape: - Almost all pathogenic bacteria are rod-shaped (Bacillus type), except filamentous (Streptomyces). Other shapes include: spherical (Cocci), ellipsoidal, spiral (Spirilla), comma-shaped (Vibrio).

Size: - Size of bacteria varies according to the spp. In young cultures bacteria range from 0.6µm -3µm length and 0.5-1.0µm in diameter. If we take a fungi spore is 50-200µm, the hyphae of fungi ranges 5-10µm in diameter and many mm in length.

Motility: - Motile or non-motile. Motile bacteria possess thread-like flagella, 10-15µm long (average 12µm long), usually longer than the cells (10times longer than the bacterial length). And about 20nm thick; flagella consist of 2-30,000 molecules of protein polymers.

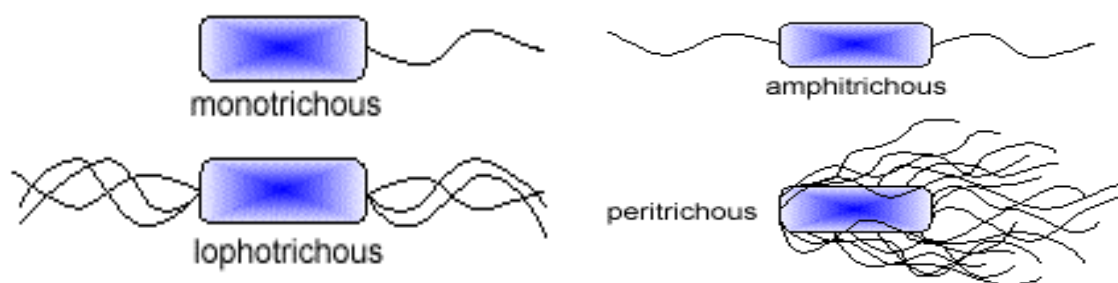
A) Atrichous: Possess no flagella on bacterial cells. E.g. The genus *Xylella* –Aflagellate, *Clavibactor*.

B) Monotrichous: with a single flagellum at the polar end. E.g. *Xanthomonas*, *Vibrio*.

C) Lophotrichous: with more than one flagellum at one or both polar ends. E.g. *Spirillum*.

D) Amphitrichous: with a tuft of flagella at each polar end. E.g. *Pseudomonas*.

E) Peritrichous: with several flagella distributed around the rod shaped bacteria. E.g. *Eriwinia*, *Agrobacterium*.



Bacterial Colony

The spherical (Cocci) bacteria may occur in different colonies:

- Diplococci – when cells occur in pairs.
- Streptococci – Occur in chains.
- Tetrad – occur in fours.
- Staphylococci – occur in irregular forms.
- Sarcinae – occur in cubical and rectangular forms.

Colony margin: colony margin may be smooth, wavy, or angular.

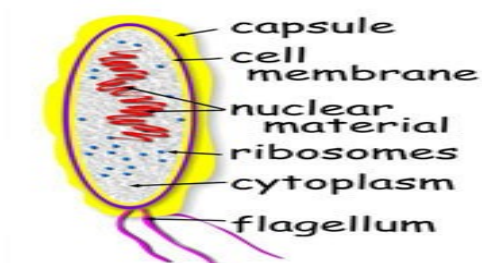
Elevation of colony: may be flat, raised, dome-shaped or wrinkled.

Colony color: whitish, grayish, but some are yellow, red or other colors.

Cytology:

Bacteria possess rigid cell-wall, cytoplasmic membrane (protoplasmic membrane), slime layer (capsule), protoplast, cytoplasm [proteins, lipids, CHO, organic cpds, minerals, H₂O, nuclear materials (chromosome), and plastid].

Below is a labelled diagram of a cell to help you identify some of these structures



Reproduction of Bacteria

Phytopathogenic bacteria reproduce by the asexual process, binary fission. Bacteria reproduce by dividing every 20 minutes. One bacterium could produce one million bacteria in 10hrs. This shows the most rapid reproduction rate. Reproduction is essential for genetic diversity through genetic recombination, but this is not the union of two gametes/no exchange of genetic material/.

☞ Three ways by which bacteria acquire new genetic material:-

1. **Conjugation:** a process by which two bacteria fuse temporarily and exchange genetic materials.
2. **Transformation:** a process in which fragments of DNA in the environment are incorporated in to a living bacterial cell to transform its genetic material.

☞ Usually these fragments of DNA found from environment dead bacteria and from the atmosphere.

3. **Transduction:** requires the involvement of two organisms (different organisms).

- Virus /bacteriophage and bacteria

☞ The virus takes genetic material bacteria (A) to which a new genetic material is incorporated to bacteria (B).

4. **Mutation:** a sudden or accidental change of genetic material b/c of various things, in this case b/c of the rapid rate of reproduction. It is the most important source of genetic diversity.

Ecology of Bacteria

Almost all plant pathogenic bacteria develops mostly in the host plant as parasites, on the plant surface, especially buds, as epiphytes and partly in plant debris or in the soil as saprophytes. Because most plant pathogenic bacteria lack specialized resistant structures to survive environmental conditions not conducive to pathogenesis, plant pathogens must coexist as epiphytes either on their host or on other plants without causing disease. Thus, endophytic growth probably represents a survival stage.

E.g. *Pectobacterium (Eriwinia) carotovorm*, which colonizes potato lenticels and cause bacterial soft rot and *Pseudomonas syringae*.

Classification of Phytopathogenic Bacteria

Taxonomy is a classification devised by humans of living organisms according to their current understanding of phylogenetic groupings. Current understanding of phylogenetics has been determined by a variety of methods, each with its own strengths and weakness. Therefore, taxonomists have strengths and weaknesses and change in response to development of the new technologies used to explore natural relationships among prokaryotes.

Classification by different criteria:

- ★ Colony color
- ★ Shape of the bacteria
- ★ Flagella –flagellate (position), aflagellate
- ★ Gram-staining test
- ★ Serological test-using antibody.
- ★ Use of selective media
- ★ Physiological requirement-
 - Salt tolerance, temperature requirement, sensitivity, etc. Thus:

Kingdom :Prokaryotae

Part I: Gram-negative aerobic rods and cocci.

Family: Pseudomonadaceae

Genera: *Pseudomonas* – Rod-shaped with one or several polar flagella Colonies: white/yellow.

Xanthomonas – Rod shaped with one polar flagellum, colonies: yellow

Family: Rhizobiaceae

Genus: *Agrobacterium* – rod shaped sparse lateral flagella. Colony: white & rarely yellow.

Genus – *Rhizobium*

Part II. Gram-negative facultative anaerobic rods

Family: Enterobacteriaceae

Genus: *Erwinia* –Peritrichous flagella; Colonies: white/yellow.

Part III. Irregular, Gram-positive, non-sporing rods. A few phytopathogenic *Corynebacterium* spp. (*Clavibacter*).

Part IV: Actinomycetes: Bacteria forming filamentous.

Genus: *Streptomyces*-Gram-positive, aerial mycelium with chains of non-motile conidia (without flagella)

2.1.3. Viruses as a pathogens

A virus is a nucleoprotein that is too small to be seen with a microscope, multiplies only in living cells, and has the ability to cause disease. All viruses are parasitic in cells and cause a multitude of diseases to all forms of living organisms, from single-celled microorganisms to large plants and animals.

Plant viruses differ greatly from all other plant pathogens not only in size and shape, but also in the simplicity of their chemical constitution and physical structure, methods of infection, multiplication, and translocation within the host, dissemination, and the symptoms they produce on the host. Because of their small size and the transparency of their bodies, viruses cannot even be viewed and detected by the methods used for other pathogens. Viruses are not cells, nor do they consist of cells.

Viruses causing diseases not by consuming cells or killing them with toxins, but by utilizing cellular substances, taking up spaces in cells, and by disrupting cellular components and processes which in turn, upset the metabolism of cells and lead to the development by the cell of abnormal substances and conditions injuries to the functions and the life of the cell. E.g. TMV.

Basic structures: viruses consist of nucleic acid (5-40%) and protein coat (capsid, 60-95%)

- ❖ Nucleic acid consists of RNA or DNA, but not both.

1. Most plant pathogen viruses consist nucleic acid with RNA.
2. Nucleic acids are the main element for infection.

- ❖ Coat proteins are proteins surrounded the nucleic acid, protecting the nucleic acid from external damage.

Infection and movements

- ❖ The infection is passive (i.e require wound or vectors)
- ❖ Movements:-follows a certain path

Reproduction:

Viruses can't reproduce by their own, but reproduce by incorporating their nucleic acid into the nucleic acid of a plant cell---interfering with normal cell division and growth of the host plants.

Classification of Viruses (Cryptogram)

Naming/grouping viruses is based on the most remarkable symptom they cause on the host. E.g. Tobacco mosaic virus (TMV). All viruses belong to:

Kingdom Vira

Division I: DNA viruses (helical/cubical)

Division II: RNA viruses (helical/cubical).

➔ 26 groups exist.

Viroids: are small, low molecular weight ribonucleic acids (RNA) that can infect plant cells, replicate themselves and cause diseases. Viroids are circular, single stranded RNA molecules. Viroids are 50 nm thick. They are associated with the cell nuclei. To date, at least 20 plant diseases have been shown to be caused by viroids.

Differences between viroids and viruses include:

1. Low molecular weight of 110,000 to 130,000 in viroids versus 10,000,000 in viruses
2. Viroids are devoid of a protein coat-free RNA.

The most important plant diseases caused by viroids:

- ✓ Potato spindle virus

- ✓ Citrus exocortis
- ✓ Tomato bunchy top
- ✓ Avocado sun blotch
- ✓ Coconut cadang-cadang

2.1.4. Nematodes as Plant Pathogens

Nematodes belong to the animal kingdom. Most of the several thousand species of nematodes live freely in fresh or salt waters or in the soil and feed on microscopic plants and animals. The nematode, which causes plant diseases are called plant nematodes. The symptoms of plant caused by plant nematode look like that caused by plant diseases. So we call it nematode diseases.

Morphology and Anatomy of Plant-pathogenic Nematodes

Plant-parasitic nematodes are small, 300-1000 μ m, with some up to 4 mm long, by 15-35 μ m long, by 15-35 μ m wide. Their small diameter makes them invisible to the naked eye, but they can be easily observed under the microscope. Nematodes are, in general, eel-shaped, and round in cross section, with smooth, unsegmented bodies, without legs or other appendages. The females of some species, however, become swollen at maturity and have pear-shaped or spheroid bodies.

The nematode body is more or less transparent. It is covered by a colorless cuticle, which is usually marked by striations or other markings. The cuticle molts when a nematode goes through its successive larval stages.

The body cavity contains a fluid through which circulation and respiration take place. The digestive system is a hollow tube extending from the mouth through the esophagus, intestine, rectum, and anus. Lips, usually six in number, surround the mouth. All plant-parasitic nematodes have a hollow stylet or spear, which used to puncture plant cells. The female nematodes have one or two ovaries, follow by an oviduct and uterus germinating in a vulva. The male is similar to the female, but there is a testis, seminal vesicle, and a terminus in a common opening with the intestine. A pair of protrusible, copulatory spicules is also present in the male.

Life cycle: Have three important steps

Egg $\Rightarrow$ Juvenile stage (larvae) $\Rightarrow$ Adult; 4 larval stages:

a. first $\Rightarrow$ second $\Rightarrow$ 1st molt

b. second $\Rightarrow$ third $\Rightarrow$ 2nd molt

c. third $\Rightarrow$ fourth $\Rightarrow$ 3rd molt

-An egg is hatched to produce a single juvenile. - 1st juvenile stage may occur in the egg.

- Each juvenile stage end/terminate with a molt. Differentiation in to males and females occur after the final molt. The life cycle from egg to egg may be completed within 2-4 weeks optimum environment if no longer in cooler temperatures.

Reproduction: Parthenogenetically or through fertilization by sperm produced in the individual (hermaphroditic) or through mating.

I. Sexual reproduction – most common.

a. **Amphigynous:** male and female are present; Zygote produce larva.

i. **Oviparous (exotikias):** eggs released on outside structures called egg sac. E.g.

Meloidogyne.

ii. **Ovoviviparity:** Fertilized egg remains inside the reproductive body of the female.

b. **Hermaphroditic:** both male and female sex organs are located on one individual.

- Females produce both sperm and eggs.

II. Parthenogenesis: eggs develop in to zygote without fertilization.

- No males are necessary.

Movement: active movement-(for only short distance) can be achieved by contraction and relaxation of muscles; passive movement: irrigation, flooding or drainage H₂O, animals feet, birds, and dust storms spread nematodes in local area, where as over long distance they are spread with farm produce and nursery plants.

Ecology of Nematodes (Mode of Life)

1. **Parasitic nematodes:** there are 30,000 spp. out of these 5000 are animal parasites ➡ 30-32

spp are parasites to human beings like ascariasis, elephantiasis, 3,000 spp are plant parasitic.

2. **Free-living:** feed on either bacteria, fungi or on soil organisms.

-Used as biological control for fungi and other pathogens.

Generally, nematodes survive in soil, plant roots, stems, kernels, etc. Abundance of nematodes 0-15cm depth; sometimes 30 -150 cm or more deep in the soil and almost all plant pathogenic nematodes live part of their lives in the soil.

Usually body made from 5 parts:

1. **Integument:** outer most covering of nematode, transparent. Consist of Cuticle, smaller glands, hypodermis etc are present
2. **Digestive systems:** made of different structures; from mouth to anus (lips, stylet, esophagus, intestine, and anus).
3. **Reproductive systems:** ovary- where eggs produced in the female, spermatheca: used to store male gametes before it fertilize the egg and, vulva: for the passage of eggs from female to outside. Male: tests, spicules, seminal vesicles, penis of protrusible.
4. **Nerve ring:** the largest connection of nematodes nerve.
- Allow the passage of information from outside and inside.
5. **Tail:** found between the anus and posterior end.

Classification of nematodes - there are different criterias:

Nematodes could be ectoparasites, endoparasites, migratory or sedentary parasites:

1. The plant parts that are attacked

A. Nematodes that attack above ground parts;

-Seed gall nematodes E.g. *Anguina*

-Leaf and bud –*Aphelenchoides*

- Attack stem - *Ditylenchus*

B. nematodes which attack underground parts

- Roots e.g. *Meloidogyne*, *Heterodera*, *Pratylenchus*

2. Mode of parasitism

a. Ectoparasites: nematodes that feed from outside. E.g. Ring nematodes (sedentary); Dagger, stubby root and sting nematodes (migratory).

E.g. Paratrichodorous, Trichodorous, *Pratylenchus*

b. Endoparasites: that is spp that enter the host and feed from within. Both of these can be:

i. **Sedentary nematodes:** which penetrate the plants but remain at one site. They don't move from one site to the other (feed at one site). E.g. *Meloidogyne* (root knot nematode), *Heterodera* and *Globodera* (Cyst nematodes).

ii. **Migratory nematodes:** Posses certain common feature to the sedentary ones but they differ by their movement from one site to the other. E.g. *Pratylenchus*: cause lesion nematodes.

3. Position of Vulva: expressed as percentage of body length measured from the anterior end and referred as the 'V' valve.

E.g. In *Ditylenchus*: the vulva located at $2/3 = 80\%$ of the body.

4. Type of Stylet: unique features of plant parasitic nematodes

- a. Stomato stylet – basal knob. E.g. *Rotylenchus*
- b. Odonostylet – no basal knob E.g. *Trichodorus*.

➔ Kingdom Animalia , Phylum Nematoda with 25 genera

Order 1: Tylenchida; Genera: 14 most important and 7 less important genera are known = 21 genera, most important parasitic genera belong to these genera.

Order 2: Dorylaimida – few belongs to this order; Genera: 4 genera known.

E.g. -Cyst nematode (*Heterodera*, *Globodera*, *Punctodera*, etc.), -Root knot nematode (*Meloidogyne* spp.), -Lesion nematode (*Pratylenchus* spp.)

Importance of Nematodes

1. They serve as biological control – by feeding on fungi, bacteria, etc.
2. Parasitic to human, animals and plants.
3. Model organisms – *C. elegans* in genetic function (study).
4. Serve as vectors for viruses, fungi, and other pathogens.
5. Nematodes make our plants susceptible to 2nd infection by fungal pathogens like *Verticillium*, *Fusarium*.

Damages by Nematodes

1. Quantitative – around 11-14% loss due plant parasitic nematodes.
2. Qualitative loss – deformation of crops, etc.
3. Biochemical damage – b/c of different enzymes and toxins.
4. Can go a disease complex -2^o infection.
5. Hinder the transfer of plant materials b/n regions.

Symptoms of Nematodes

- ★ On roots: root knot/goals by *Meloidogyne* spp.
- root lesions by *Pratylenchus*.
- Excessive root branching.

- Thickening of root tips.

★ Above ground parts: -stunting, yellowing, wilting.

- ✓ Necrotic lesions.
- ✓ Galls on stem.
- ✓ Seed galls.

Control measures

1. Cultural practices – selection of appropriate site, planting time, crop rotation, etc.
2. Resistant varieties/tolerant varieties,
3. Hot water treatment ~45°C for 2hrs.
4. Use of trap crops.
5. Use of chemicals, Nematicides.

2.1.5. Parasitic flowering plants as pathogens

Several higher plants, which produce flowers and seeds, live as parasites on other plants (host plants), deriving their nutrition from the host plants and adversely affecting their growth and yield. The most common example under our condition is dodder (also known as strangle weed, devil's hair), which parasitizes several plants like potatoes, sugar beets, alfalfa and carrots. Dodder produces orange or yellow vine strands that entwine the stems and other plant parts from which it draws its nutrition through tube-like structures it introduces into the host tissue. Dodder produces abundant number of seeds which ensure its propagation and spread. Another example of a parasitic plant is dwarf mistletoe on pines.

2.2. Non-infectious causes of plant diseases

Almost any of a variety of the environmental and cultural factors, under extreme or improper conditions, can cause disease in plants. Since these diseases occur in the **absence** of pathogens, they do not spread from diseased plant to healthy plant.

2.2.1. Temperature

When plants or plant parts are exposed to high temperatures for prolonged periods, symptoms of scorching or scalding may develop on the exposed parts. Some examples: sunburn or scorching of leaves, sunscald of fruits (example: apples, tomatoes, peppers, melons). Similarly, low temperatures like frost or freeze, can damage the exposed or sensitive organs (leaves, buds,

flowers, young fruits, etc.) or may kill the entire plant. Some examples: southwest side damage to trunks of apple trees, frost damage to blossoms and young apple fruits, russet ring (caused by frost) on apple and pear fruits, winter injury to trees, frost damage to tomatoes, beans, potatoes, etc.

2.2.2. Moisture

Too much soil moisture, due to excessive watering, poor drainage, flooding, may cause the plants to turn yellow and stunted. The plants may die due to lack of aeration to the root system or due to root rots and damping-off caused by pathogens that thrive under these conditions. Some examples: poor development and root rots of potted indoor plants, damping-off of seedlings. In some indoor or greenhouse plants (example: geraniums, begonias) growing under warm, humid atmosphere and excessive soil moisture, a condition known as edema (small, wart-like, rusty, corky bumps) may develop on the underside of the leaves, and on stems. At the other extreme, low moisture availability or drought will lead to poor development, wilting and death of the plants.

2.2.3. Nutrition

Plants require several major elements (nitrogen, phosphorus, potassium, calcium, magnesium and sulfur) and minor elements (iron, boron, copper, zinc, etc.) for normal growth. Deficiency (absence or inadequate amount) of any of these essential nutrients results in disease symptoms in the plant, specific symptoms being dependent on the plant species and the deficient nutrient. If not corrected, prolonged acute deficiency of essential nutrients may lead to death of the plant. Some examples: nitrogen deficiency in beans; iron deficiency in many fruit trees and landscape plants; zinc deficiency in apple trees; bitter pit of apple fruit caused by calcium deficiency; blossom end rot of tomato fruit caused, in part, by calcium deficiency; cracking of pear fruits due to boron deficiency.

2.2.4. Improper cultural practices

Any of the cultural practices done in a wrong way or at the wrong time may result in significant damage to the plants. Injury due to excessive amounts of chemical (fertilizer or pesticide) or

improper chemical mixes in the spray tank, excessively deep cultivation leading to root pruning, distorted and twisted roots of pot-bound plants, are all examples of such a type of damage. African violet leaves sprinkled with very cold water develop rings and ring-like patterns that resemble symptoms caused by some viruses.

2.2.5. Pesticide Toxicity

Some of the pesticides, if improperly used, may cause serious damage to the plants. For example, if wettable sulfur is sprayed (for powdery mildew control) on a very hot day (above 90°F), it will result in injury to the plant (phytotoxicity). However, the most common type of chemical injury to the plants in recent years has been due to soil residues or spray contamination or drift of herbicides on to non-target plants. Examples: 2,4-D damage to beans and tomatoes, dicamba (Banvel) damage to vegetables and trees, glyphosate (Roundup) damage to fruit trees. Some of the persistent herbicides used as soil sterilants may leave the ground unsuitable for any plant growth for several years.

2.2.6. Unfavorable Soil pH

Although many plants can grow in a rather wide range of soil pH, plants growing in soils with unfavorable pH (too acidic or too alkaline) usually show poor growth and mineral deficiency or toxicity symptoms. Example: Iron deficiency symptoms are very common in plants growing in high pH soils in Idaho. Under conditions of high soil pH, iron in the soil becomes unavailable to the plant, thus inducing interveinal chlorosis and yellowing of leaves followed by twig die-back. The plant may die if the condition remains uncorrected for prolonged time.

Chapter 3

KOCH'S POSTULATES AND DISEASE SYMPTOMS

3.1. Koch's Postulates (Rules) – Procedures

The hundred years following de Bray's first plant disease identification were very active in the area of new disease diagnosis. The procedure used to confirm the existence of a new plant disease

was developed by Anton Koch in 1876 and is known as Koch's Postulates or Proof of Pathogenicity. This procedure is still used today for the diagnosis and identification of a new (previously unreported) plant disease. The procedure involved the following four steps.

1. **Association:** The suspected pathogen must be consistently associated with the diseased plant (or animal).
2. **Isolation:** The pathogen must be isolated and grown in pure culture and its characteristics described.
3. **Inoculation:** The pathogen from pure culture is inoculated into a healthy plant of the same species or variety and it must produce the same symptoms and signs.
4. **Re-Isolation:** The pathogen is re-isolated from the inoculated plant and its characteristics must be the same as the organism initially isolated in step 2.

3.2. Morphological Symptoms of Plant Diseases

Symptoms refer to manifestations of physiological reactions of plants to the harmful activities of pathogens. Disease symptoms are of two types:

1. Morphological symptoms externally detectable.
2. Histological symptoms (pathological anatomy) – detectable only by microscopic examinations.

Morphological symptoms - Necrosis

- Hypoplasies
- Hyperplasies

3.2.1. Necrotic Symptoms

Necrosis: - This term is used to indicate the condition in which the death of cells, tissues, and organs has occurred as a result of the parasitic activity. The characteristics appearance of the dead area differs with different hosts and host organs and with different parasites so that there are different types of necrotic symptoms. E.g. spots, streaks, stripe, canker, blight, damping-off, burns, scald or scotch and rot are results of necrosis of tissues.

3.2.2. Hypoplastic Symptoms (Atrophy) or dwarfing: reduction in the size or number of cells.

In many diseases one of the results is inhibition of growth resulting in stunting or dwarfing. The

whole plant may be dwarfed or only certain organs may be so affected. Sometimes hypertrophy and atrophy both are present in the same organ. Hypoplasia refers to the underdevelopment of cells, tissues or organs due to decreased cell division or slow growth. E.g. dwarfing, resetting (absence of internode elongation and clustering of leaves). Chlorosis (etiolation) is also another example.

3.2.3. Hyperplastic Symptoms (Hypertrophy) –overgrowth: due to cell division

-Hyperplasia is the abnormal increase in size of a plant organ due to an increase in number of cells of the organ.

-Hypertrophy: the increased size of the organ is due to increase in the size of cells (enlargement).

Both the conditions may be simultaneously present. In some diseases the increase in size of the plant organs is due to increase in size of cells and also due to presence of fungus structures. E.g. Galls, knots, leaf curl, pockets and bladders, withes' broom, and hairy root are all the results of some form of overgrowth.

-Is due to overdevelopment in size or color of plants or plant organs, or the abnormal early development of plant organs. This could be due to an excessive increase in size of a cell, a tissue, or an entire plant. E.g. galls, root knots, tumor.

5.3. Effects of Plant Pathogens on Plant Physiology

The effects include:

1. Effects on photosynthesis
2. Effects on translocation
3. Effects on respiration
4. Effects on cell permeability
5. Effects on translation and transcription etc.

3.4. Diagnosis of Plant Diseases

Disease diagnosis is the process of disease identification.

★ The first and most important step in managing a plant disease is to correctly identify it. Although some diseases can be diagnosed quickly by visual examination, others require laboratory testing for diagnosis. These laboratory procedures may take days or even weeks to complete and are, in

some cases, relatively insensitive. Delays are frustrating when a quick diagnosis is needed so that appropriate disease control measures may be taken to prevent plant injury.

- ★Plants should be initially examined in the field, landscape, or garden setting, and the site, plant and problem history should be determined with experience and constant study.
- ★ It is the process of biotic or abiotic disease identification.
- ★The diagnosis of plant diseases is a scientific art that is improved with experience and constant study.
- ★As diagnosticians we need practical knowledge of the host and its growing conditions as well as pertinent scientific background (techniques and principles).
- ★Plant disease diagnosis is designed to recognize the primary disease-causing agents.
- ★An effective diagnostic test should be as:-
 - ✓ Simple, Accurate, Rapid and, Safe to perform as the technology allows, yet also be sufficiently sensitive to avoid false positives.

Steps in Disease Diagnosis:-

The steps in routine (practical) disease diagnosis are:-

1. Obtain background information on host and diseases
2. Obtain a good sample of a diseased plants
3. Examine plants and describe symptoms and signs of disease
4. Obtain literature describing diseases of the host
5. Identify disease by comparing your description of disease with published descriptions

Further steps in disease diagnosis:

6. If the above steps do not lead to a positive diagnosis of the disease, isolation and identification of the pathogen may be necessary.
7. In some cases experimental reproduction of the disease may be required and rules for proving the cause of a disease popularly known as “Koch’s Postulates” are used.

-Diagnosis requires a blending of science, experience, observation and art.

-It is a cooperative effort between the growers and the plant expert.

Chapter 4

Host-parasitic relations/ parasitism and disease development

4.1. Host-parasitic relations/ parasitism

Types of interactions

- ❖ **Parasitism:** one organism taking its nourishment from another, and in doing so causing disease
- ❖ **Mutualism:** two organisms living together, in close association, with both benefiting from the interaction. Eg. Rhizobium on legumes, nitrogen fixation, lichens (fungi + algae), some plant endophytes.
- ❖ **Commensalism:** two organisms living together, where one derives benefit and the other is unaffected (+,0)

Types of organisms

- **Symbiont:** an organism involved in a sym-biotic relationship.
- **Saprophyte:** an organism that derives its nourishment from dead organic matter.
- **Parasite/pathogen:** an organism that de-rives its nourishment from another, and in doing so causes disease.

Types of parasites

- **Obligate parasite:** a parasite that can grow only in association with a living host.
Also called **biotrophs**.
- **Non-obligate parasite:** an organism that can survive either as a parasite or a saprophyte.
- **Facultative parasite:** an organism that can be a parasite under the appropriate conditions (but is primarily a saprophyte).
- **Facultative saprophyte:** an organism that can be a saprophyte (but is primarily a parasite).

4.2. Disease Development in plants

It is important to understand how plant diseases develop in order to control them. By the time it becomes obvious that a plant has a disease, it is generally too late to do anything about it in that

growing season. Plants cannot be cured in the way people expect their own ills to be cured. The amount of disease that develops in a plant community is dependent on properties of the host, the pathogen and the environment.

The environment can affect both the susceptibility of the host (e.g. by creating stress in the plant) and the activity of the pathogen (e.g. providing moisture for spore germination). The pathogen and the host can affect each other's performance. The plant can also change its environment, by creating a microclimate around it. The relationship of these factors is called the disease triangle. If only a part of the triangle exists, disease will not occur. Understanding the disease triangle helps us understand why most plants are not affected by the many thousands of diseases that exist.

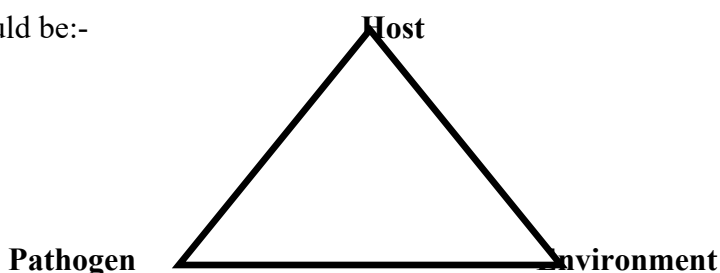
Table.1. Factors that affect disease development.

Pathogen	Host	Environment
• Presence of pathogen • Pathogenicity • Adaptability • Dispersal efficiency • Survival efficiency • Reproductive fitness	• Susceptibility • Growth stage & form • Population density & structure • General health	• Temperature • Rainfall / Dew • Leaf wetness period • Soil properties • Wind • Fire history • Air pollution • Herbicide damage

4.3. Disease Triangle: - a triangular representation of the prerequisite of plant disease. It is a triangle that depicts (shows) the pre-requisites for disease development. The length of each side is proportional to the sum total of the characteristics of each component that favors a disease.

Any plant disease to be occur, there should be:-

- Host (susceptible host)
- Pathogen (at virulent stage)
- Environment



Examples

1. Fusarium head blight

There should be –wheat at flowering

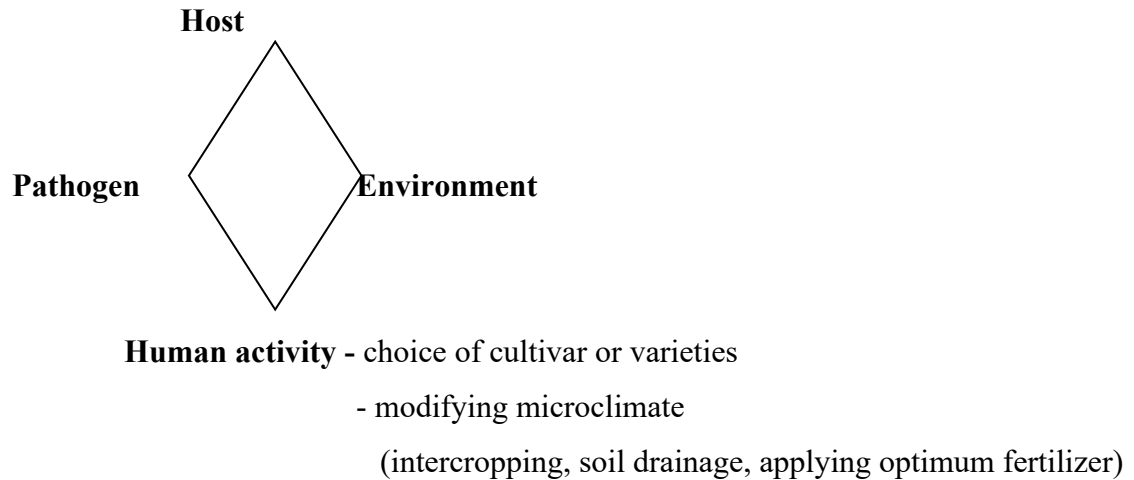
- _ spore of the fungi (fusarium)
- _ Humid and warm temp. (18-25⁰c)

2. Late blight of potato

There should be –potato/ tomato at flowering/vegetative stage

- _ spore of the fungi (*P. infestance*)
- _ Cool and humid

The disease triangle has been modified into:-Disease tetrahedral

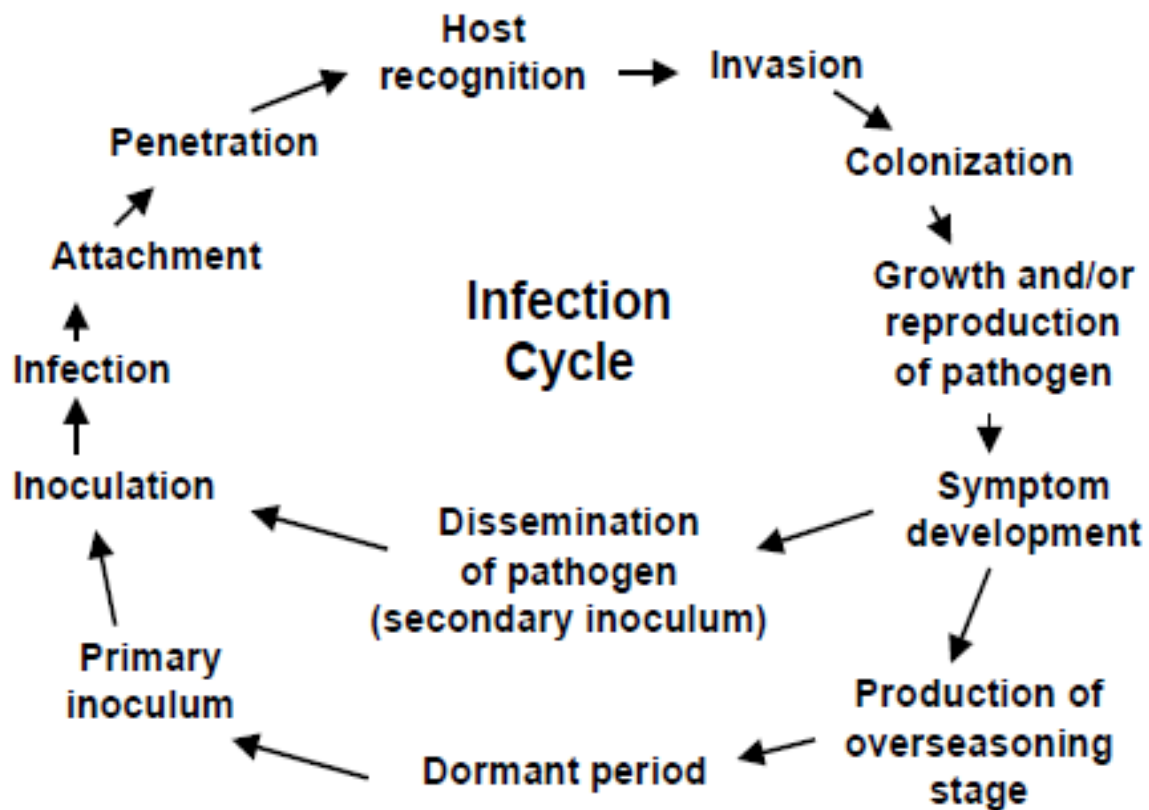


* **Effective disease** control measures aim at breaking this environment host pathogen triangle.

4.4. Stages in disease development

In every infectious disease there is a series of more or less distinct events occurs in succession and leads to the development and perpetuation of the disease and the pathogen. This chain of events is known as **Disease Cycle**.

The primary events in a disease cycle are inoculation, penetration, establishment of infection, colonization (invasion), growth and reproduction of the pathogen, dissemination of the pathogen, and survival of the pathogen in the absence of the host.



4.4.1. Inoculum and Inoculation

Inoculation: the coming in contact of a pathogen or its inoculum with the host.

Inoculum: any part of a pathogen that imitates infection. E.g. spores, sclerotia, fragment of mycelium of fungi or whole individuals of bacteria, viruses, protozoa, nematode eggs, larva, or adults.

There are two types of inoculum:

1. **Primary inoculum:** responsible to cause initial (1^0) infection on overwintering season.
2. **Secondary inoculum:** results of pathogen multiplication after primary inoculum. It causes 2^0 infection infects that either previously infected plants or healthy plants.

Inoculum potential (IP): genetic ability of the inoculum to cause disease.

Inoculum density (ID): number of inocula lands on host.

❖ The severity of the disease is directly proportional to IP and ID; that is $\text{disease severity} = \text{IP} \times \text{ID}$.

Inoculum: - parts of the pathogen that initiates infection.

- Fungi.....spore, mycelium
- Bacteria & virus.....the whole organism
- Nematodes....eggs, larvae, Adult
- Parasitic flowering plant.....Seeds

Source of Inoculum:

- Plant debris, or soil in the field where the crop is grown, seed, transplants, tubers, propagative materials (cuttings, etc.).
- Nearly infected plants, perennial weeds or alternate hosts (volunteer plants, left-over plants, ratoons, collaterals, etc). Inocula may live on or in the host.

4.4.2. Pre-penetration and Penetration Process

Pre-penetration Process: includes: germination of fungal spores and seeds of parasitic plants (e.g dodder), hatching of nematode eggs, pathogen anchorage through haustoria's (e.g. parasitic plants), host pathogen recognition achieved by chemicals that are secreted by host pathogens.

Pathogen penetration: entrance of a pathogen in to its host.

☞Pathogens penetrate host plants through various mechanisms:

Two kinds of penetration

A. direct Penetration: that usually employed by:

- ✓ Fungi: produce hydrolytic enzymes, use spores.
Spore→germ tube→haustoria (cushion) →Infection peg.
- ✓ nematodes: use stylet and make hole in the host.
- ✓ Parasitic plants use haustoria's.

B. Indirect Penetrations:

1. through natural openings: stomata, hydathodes, nectarthodes, lenticles. E.g. Fungi.
 2. Wounds: all bacteria, most fungi, some viruses, all viroids, MLOs and RLOs.
- ☞Viruses, viroids, MLOs and RLOs penetrate via wounds produced by vectors.

4.4.3. Infection and Establishment

Infection: a process by which the pathogen establishes contact with susceptible cells and tissues of the host and procures nutrients from them.

Two types of Infection:

- i. **Active:** able to produce disease symptoms immediately after inoculation.
- ii. **Latent:** not able to produce symptoms right away.

Incubation Period: The time intervals between inoculation and symptom development. The pathogen is multiply during this time. This depends on the pathogen-host combination, stage of host, temperature, etc.

Invasion: The spread of a pathogen within the host. Most pathogens spread in to all the tissues of the plant organs (roots, leaves, stems, and fruits). The invasion could be local or systemic. E.g. wilt fungi in xylem vessels (systemic disease).

4.4.4. Reproduction of pathogens

Fungi reproduce by spores (cryptogamy), spores could be produced on or in host tissues; fungi also grows by means of mycelium (endophytic or epiphytic); parasitic seed plants by seeds; bacteria, MPLOs, and protozoa by binary fission; viruses duplicate in cells; nematodes reproduced through eggs.

4.4.5. Dissemination/spread of pathogens

Autonomous movement by pathogens is limited. Only a few pathogens move in this way. E.g. nematodes; fungal spores, hyphae, rhizomes; bacteria (flagella); Parasitic plants and fruits through shattering.

Passive movement occurs through: air, water, insects, humans and animals, mites, nematodes and other vectors.

4.2.6. Survival of Pathogens

***Pathogens can survive** –as mycelium in tissue

-as spores on plant surface and bud scales, on perennial hosts, on plant debris, fruits, seed, and leaves;

- as mycelium, common spores, resting spores (chlamydospores), sclerotia, seeds and tubers, in the soil.

Soil borne pathogens (soil invaders) can be:

- a) Soil transients (live for shorter period)
- b) Soil inhabitants (live for longer period)

*nematodes as eggs in soil, adults (larvae) in plant tissues or debris, seeds, bulbs for many years

*Parasitic higher plants as seeds in soil, as vegetative form on the hosts. E.g. dodders.

Chapter 5

Defence mechanisms of plants against pathogens

5.1. Mechanisms of Attack by Pathogens/ How pathogen attack plants

Why do Pathogens Attack plants?

Pathogens attack plants because during their evolutionary development they have acquired the ability to live off the substances manufactured by the host plants, and some of the pathogens depend on these substances for survival.

Therefore, for a pathogen to infect a plant it must be able to make its way into and through the plant, obtain nutrients from the plant, and neutralize the defense reactions of the plant.

5.1.1. Weapons of pathogen attacks

❖ Mechanical forces

Plant pathogens are, generally, tiny microorganisms that cannot apply a “voluntary” force to a plant surface. Only some fungi, parasitic higher plants, and nematodes appear to apply mechanical pressure to the plant surface they are about to penetrate. The amount of pressure, however, may vary greatly with the degree of “pre-softening” of a plant surface by enzymatic secretions of the pathogen.

- **Fungi** use the aspersorium (club like structure developed at the tip of germ tube of fungi)
- **Nematodes** penetrate plant surfaces by means of the stylet, which is thrust back and forth and exerts mechanical pressure on the cell wall
- **Parasitic flowering plants** use their haustoria

❖ Chemical weapons

Although some pathogens may use mechanical force to penetrate plant tissues, the activities of pathogens in plants are largely chemical in nature. Therefore, the effects caused by pathogens on plants are almost entirely the result of biochemical reactions taking place between substances secreted by the pathogen and those present in, or produced by, the plant.

Chemicals produced by pathogen used to facilitate the penetration and to invade. The main groups of substances secreted by pathogens in plants that seem to be involved in production of disease, either directly or indirectly, are enzymes, toxins, growth regulators, and polysaccharides.

These substances vary greatly as to their importance in pathogenicity, and their relative importance may be different from one disease to another. Thus, in some diseases, such as soft rots, enzymes seem to be by far the most important, whereas in diseases such as crown gall, growth regulators are apparently the main substances involved. Among the plant pathogens, all except viruses and viroids can probably produce enzymes, growth regulators, and polysaccharides

A. Enzymes

Enzymes are generally large protein molecules that catalyze organic reactions in living cells and in solutions.

- ✓ Facilitate chemical reaction (biological point of view)
- ✓ Important to break the defense mechanism

Eg. Cellulase, Glucanase, Xylanase, Amylase, Protase

❖ Enzymes can be produced by both the plant and the pathogen

- If by pathogen it disintegrate the plant defense mechanisms
- If by plant it enhance the defense mechanism

B. Toxins

Are low molecular weight produced by pathogens. Toxins act directly on living host protoplasts, seriously damaging or killing the cells of the plant. Some toxins act as general protoplasmic poisons and affect many species of plants representing different families known as **non-host specific toxins**. Others are toxic to only a few plant species or varieties and are completely harmless to others known as **host specific toxins**. Many toxins exist in multiple forms that have

different influence. Are poisonous that leads to cellular death and are important in pathogenesis because:-

- They affect the permeability of cell membrane thus block the movement of materials.
- Inhibit or inactivate enzymes of the host
- Cause deficiency of hormones => finally leads to the death of cells.

i. Non-host specific toxins

- ✓ Toxins produced by pathogens and that affect a wide range of host plants (i.e no preference)

These toxins increase the severity of disease caused by a pathogen, i.e., they affect the virulence of the pathogen, but are not essential for the pathogen to cause disease, i.e., they do not determine the pathogenicity of the pathogen.

ii. Host specific toxins

A host-specific or host-selective toxin is a substance produced by a pathogenic microorganism that, at physiological concentrations, is toxic only to the hosts of that pathogen and shows little or no toxicity against non susceptible plants.

- ✓ Known by acting on specific host

Some examples of toxin

Toxin	Pathogen	Host
Host specific (selective)		
Victorin (HV-toxin)	<i>Helminthosporium (Cochliobolus) victoriae</i>	Oats
T-toxin	<i>H. maydis (C. heterostrophus)</i>	Maize
HC-toxin	<i>H. (C.) carbonum</i>	Maize
HS-toxin	<i>H. (C.) sacchari</i>	Sugarcane
PC-toxin	<i>Perioconia circinata</i>	Sorghum
Ptr-toxin	<i>Pyrenophora tritici-repentis</i>	Wheat
AK-toxin	<i>Alternaria kikuchiana</i>	Pear
Host non-specific (non selective)		
Fusicoccin	<i>Fusicoccum amygdali</i>	Almond
Cercosporin	<i>Cercospora</i> spp.	Various
Tentoxin	<i>Alternaria</i> spp.	Various
Tabtoxin	<i>Pseudomonas syringae</i> pv. <i>tabaci</i>	Tobacco
Phaseolotoxin	<i>P.s.</i> pv. <i>coronafaciens</i>	Cereals
Syringomycin	<i>P.s.</i> pv. <i>phaseolicola</i>	Bean
Coronatine	<i>P.S.</i> pv. <i>syringae</i>	Various
	<i>P.S.</i> pv. <i>glycinae</i> and others	Soybean

C. Growth regulators

The most important growth regulators are **auxins**, **gibberellins**, and **cytokinins**, but other compounds, such as ethylene and growth inhibitors, play important regulatory roles in the life of the plant. Growth regulators act in very small concentrations and even slight deviations from the normal concentration may bring about strikingly different plant growth patterns.

They are important for normal growth and development if and if it is present at a normal level. But sometimes the pathogen induce these hormone to produce beyond the optimum level and result abnormal growth like stem malformation, extensive rooting and branching

- ✓ **Plant pathogens may produce more of the same growth regulators as those produced by the plant**

D. Polysaccharides

- ✓ Pathogens like fungi, bacteria, nematodes constantly release varying amounts of polysaccharides
- ✓ Are mucilaginous substance which are important to produce a better interface between host and pathogen
- ✓ Important to facilitate pathogenesis because it ensure pathogen survival and promote colonization
- ✓ Particularly important in wilt diseases caused by pathogens that invade the vascular system of the plant.

E.g. *Fussarium oxysporium* f.sp. *lycoprisci*

- released large polysaccharide molecules in the xylem and cause a mechanical blockage of vascular bundles and thus initiate wilting

5.2. How plant defend themselves against pathogen

Plants defend themselves against pathogens by a combination of two weapons forms: (1) Structural characteristics that act as physical barriers and inhibit the pathogen from gaining entrance and spreading through the plant. (2) Biochemical reactions that take place in the cells and tissues of the plant and produce substances that are either toxic to the pathogen or create conditions that inhibit growth of the pathogen in the plant. The combinations of structural characteristics and biochemical reactions employed in the defense of plants are different in different host–pathogen systems. In

addition, even within the same host and pathogen, the combinations vary with the age of the plant, the kind of plant organ and tissue attacked, the nutritional condition of the plant, and the weather conditions.

5.2.1. Structural Defense Mechanisms

5.2.1.1. Pre-existing defense structures (passive mechanisms)

These are defense structures that are present in the present even before the pathogen is inoculated. These include:

- a) Amount and quality of wax and cuticle covering epidermis
- b) Structure of structure cell wall
- c) The size, location and shapes of stomata and time of stomata opening
- d) Thickness and toughness of cell walls

Wax and trichomes of leaf and fruit surfaces repel film of water that may be required for germination and multiplication of pathogens. Thick cuticles may prevent penetration by pathogen. Stomata that may open late in the day or very narrow stomata or broad but elevated stomata surrounded by guard cells prevent penetration.

5.2.1.2. Post-infectional defense structures (Active Mechanisms)

Plants may defend themselves against pathogens even after penetration by pathogens. This could be by forming one or more types of structures that prevent further pathogen invasion. These include:

- a) Histological defense structure
- b) Cellular defense structures
- c) Cytoplasmic structures
- d) Necrotic/ hypersensitive structures

A. Histological defense structure

i. Formation of cork layers

By inducing plants to form several layers of cork cells beyond the point of infection by stimulating host through secretion of pathogens. The layer block spread of toxins and of pathogens.

ii. Formation of abscission layers

This is formed on young and active layers of stone fruit trees after infection by any of several fungi, bacteria or virus.

iii. Formation of Tyloses

These are formed in xylem vessels of most plants during infection by vascular pathogens. They are overgrowths of paranchymatous cells and clog the vessels, block advancement of pathogens.

iv. Deposition of gums

Different gums forms around infection sites and deposited in the intercellular spaces and within the cells, serving as impenetrable barrier and enclose the pathogens.

B. Cellular Defense Structures

This is the result of morphological changes in the cell wall of the invaded cell by pathogen. The mechanisms involved here include:

- i. Swelling of the outer layer of the cell wall of parenchyma cell producing fibrillar material that traps bacteria and prevents multiplication
- ii. Thickening of cell wall in response to viral and fungal infection. Phenolic compounds may be produced too.
- iii. Callose papillae deposition on the inner side of cell walls in response to fungal infection => Prevents penetration by pathogens.

C. Cytoplasmic Defense Reaction

Here the cytoplasm surrounds the clump of hyphea and the nucleus is stretched to the point where it breaks in two. The cytoplasm becomes granular and dense, various structures appear. Eventually the pathogen disintegrates.

Necrotic Defense Reaction

When the pathogen penetrates the cell wall, the nucleus approaches the pathogen in the protoplast; it disintegrates produces brown resin-like granules in the cytoplasm, the cell disintegrates and pathogen degenerates. Eg. By obligate fungal parasite, viruses and nematode infection.

5.2.2. Biochemical Defense Mechanisms

5.2.2.1. Pre-existing biochemical defense (Passive Resistance)

a. Inhibitors/ Fungitoxic Exudates Released into Plant Environment

Inhibitors of pathogens are secreted by the plants and exuded through the plant surface.

b. Inhibitors prevent in plant cells before infection.

Several Phenolic compounds have been proposed to be present in high concentrations in cells of young fruits and leaves. As the plant becomes older, the contents of the inhibitors decrease and resistance lowers too.

c. Lack of Essential Factors

These include:

- i. Lack of recognition between host and pathogen
- ii. Lack of host receptors and sensitive sites for toxins
- iii. Lack or low concentration of essential nutrients for pathogen

5.2.2.2. Induced Biochemical Defense (Active Resistance)

This is an active resistance mechanism and is performed in any one or more of the following responses

a) Biochemical inhibitors produced in plants due to infection

When plants cells and tissues are injured by pathogens, chemical against or mechanical means, they produce fungitoxic substances around the sites of injury such as *chlorogenic* and *caffeic* acids, oxidation products of Phenolic compounds and *phytoalexins* => inhibit the growth of fungi and bacteria.

b) Hypersensitive reaction

This occurs in compatible combinations of hosts and pathogens (fungi, bacteria, viruses and nematodes). The pathogens within the operation area are isolated by necrotic tissue and die quickly; thus the pathogen spread is restricted.

c) Increase in Phenolic compound concentration

d) Formation of substances resisting pathogen enzymes

Complexes between pectin and proteins and polyvalent cations and magnesium near the infection sites result in formation of pectin salts or other complexes that resist degradations by the pathogen enzymes => prevent tissue maceration and restrict pathogen.

e) Inactivation of pathogen enzymes

Polyphenols have inhibitory action on pectolytic and pectinolytic enzymes of the pathogen. Eg. Resistance of apple to *Monilinia*, grape to *Botrytis*.

f) Detoxification of pathogen Toxins

Detoxification of toxins such as fusaric acid pyricularin is known to occur in plants through metabolization or combination with other substances and formation of less toxic or non-toxic compounds.

g) Induced Resistance

In plants induced resistance is a resistance type that develops after pre-inoculation treatment of plants with various biotic agents or after pre-treatment with various chemical or physical agents. It is a non-specific resistance against fungi, bacteria, various and even insects.

Chapter 6

Genetics of Plant Diseases

6.1. Variability in organisms

All individuals produced as a result of a sexual process, such as the children of one family, are expected to be different from one another and from their parents in a number of characteristics, although they retain most similarities with them and belong to the same species. This is true oomycetes and of fungi produced from sexual spores such as oospores, ascospores, and basidiospores; of parasitic higher plants produced from seeds; and of nematodes produced from fertilized eggs.

When individuals are produced asexually, the frequency and degree of variability among the progeny are reduced greatly, but even then certain individuals among the progeny will show different characteristics. This is the case in asexual reproduction of fungi by means of conidia, zoospores, sclerotia, and uredospores, and in bacteria, mollicutes, and viruses.

6.2. Mechanism of variability

There are two main categories of mechanisms of variability:

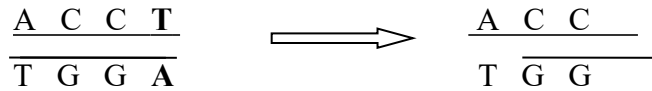
- I. General mechanisms of variability
- II. Specialized mechanisms of variability

I. General mechanisms of variability

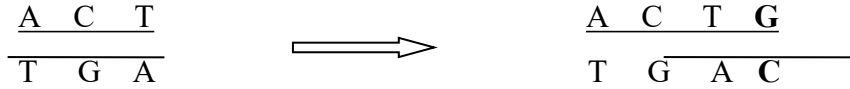
A. Mutations

Mutation is a more or less unexpected change in the genetic material of an organism, which is then transmitted to the progeny in a hereditary fashion. Mutations occur spontaneously in nature in all living organisms that reproduce through: (1) only sexually; (2) only asexually; and (3) both sexually and asexually. Actually, mutations represent changes in the sequence of bases [Adenine-A, Guanine-G, Cytocine_C.Thymine-T and Uracil-U] in the DNA **through:**

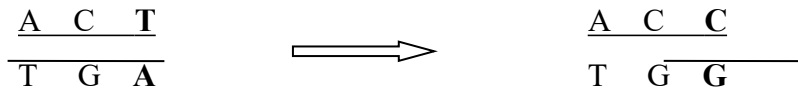
I. Deletion of some parts of an organisms



II. Addition of another genetic materials



III. Substitution genetic materials



Factor favoring mutations includes:

- Great number of progeny produced by pathogens
- Monocultural cropping system and
- Planting only a few genetically homogenous varieties of each crop continuously over enormous land expanses for a number of years.

B. Recombination

Recombination occurs during the sexual reproduction of plants, fungi, and nematodes whenever two haploid (1N) nuclei, containing genetic material that may differ in many loci, unite to form a diploid (2N) nucleus, called a zygote.

- Works for those pathogen that can reproduce sexually

II. Specialized mechanisms of variability in pathogens

These specialized mechanisms of variability include heterokaryosis, heteroploidy, and parasexualism in fungi; conjugation, transformation, and transduction in bacteria; and genetic recombination in viruses.

A. Sexual-like processes in fungi

1) Heterokaryosis

It is the condition in which, as a result of fertilization or anastomosis, cells of fungal hyphae or parts of hyphae contain two or more nuclei that are genetically different.

It is common phenomena in Basidiomycetes in which the dikaryotic state may differ drastically from the haploid mycelium and spores of the fungus.

For example in *P.graminis tritici*,

- basidiosporescan infect barberry but not wheat
- aeciospores and uredosporescan infect wheat but not barberry
- dikaryotic mycelium..... can grow in both barberry and wheat

2) Heteroploidy

It is the existence of cells, tissues, or whole organisms with numbers of chromosomes per nucleus that are different from the normal 1N or 2N complement for the particular organism. Heteroploids may be haploids, diploids, triploids, or tetraploids or they may be aneuploids, i.e., have one, two, three, or more extra chromosomes or are missing one or more chromosomes from the normal euploid number.

3) Parasexualism

It is the process by which genetic recombination can occur within fungal heterokaryons. This occurs by the occasional fusion of the two nuclei and the formation of a diploid nucleus (two chromosomes). During multiplication, crossing-over occurs in a few mitotic divisions and results in the appearance of genetic recombinants by the occasional separation of the diploid nucleus in to its haploid components.

B. Sexual-like processes in Bacteria

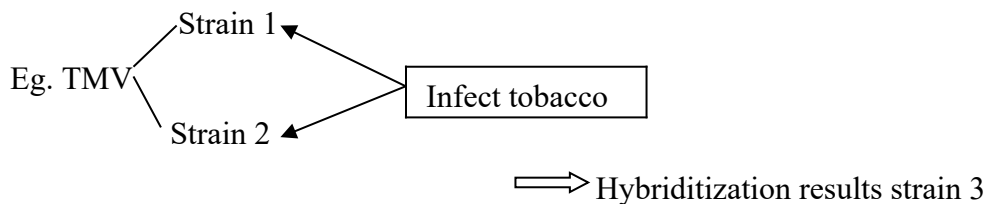
There are three sexual-like processes in bacteria:

2. **Conjugation:** two compatible bacteria come in contact with each other and small portion of the chromosomal or non-chromosomal (plasmid) genetic material of one bacterium is transferred to the genetic material of the other.
3. **Transformation:** bacterial cells are transformed genetically by absorbing and incorporated in to their own cells genetic materials secreted by or released during rupture of other compatible bacteria.
4. **Transduction:** a bacterial virus (bacteriophage) transfers genetic material from the bacterium in which the phage was produced to the bacterium it infects next.

C. Genetic Recombination in Viruses

There are two ways

1. Mutation
2. Hybridization: the kind of reproduction between two kind of virus strain which infect the host at the same times.



6.3. Stages of Variation in Pathogens

The entire population of a particular organism on the earth, e.g., a fungal pathogen, has certain morphological and other phenotypic characteristics in common and makes up the **species** of pathogen, such as *Puccinia graminis*, the cause of stem rust of cereals. Some individuals of this species, however, attack only wheat, barley, or oats, and these individuals make up groups that are called **varieties** or **special forms** (*formae specialis*) such as *P. graminis* f. sp. *tritici* or *P. graminis tritici*, *P. graminis hordei*, and *P. graminis avenae*.

Class--- Basidiomycetes

Order---Uredinales

Genus---Puccinia

Spp.----*Puccinia graminis*

F.sp.---- *P. graminis f.sp.tritici*

Races----R1, R2, R3 with different genetic makeup which is important to break the resistance ability of the plant.

Bacteria

Genus → Spp. → Pathovar → Strain

To determine the identity of Races

- Collect isolates of the pathogen
- Grow different varieties in protected environments
- Inoculation
- Check for the host reaction

Eg.

Varieties	Isolate of pathogens				
	1	2	3	4	5
A	+	+	+	—	+
B	—	+	—	—	+
C	—	+	—	—	+
D	+	+	+	—	+
E	+	+	+	—	—
F	+	+	+	—	—
G	+	+	+	+	—

Isolate 1 and 3 are the same thus there are four races

We can also identify the most and the least virulent isolates (race)

- ✓ Race 2 ----the most virulent
- ✓ Race 4.....the least virulent

6.4. Genetics of Virulence in Pathogens

Infectious plant diseases are the result of the interaction of at least two organisms, the host plant and the pathogen. The properties of each of these two organisms are governed by their genetic material, the DNA, which is organized in numerous segments making up the genes.

Gene-for - gene concept: for each gene that confers resistance in the host there is a corresponding gene in the pathogen that confers virulence to the pathogen and vice versa. See the following quadratic check:-

Genes of the pathogen	Genes in the plant	
	R	r
A	AR (Resistant)	Ar (Susceptible)
a	aR (Susceptible)	ar (Susceptible)

Key : A= represents the dominant gene for avirulence (non-pathogenicity) and

‘a’ = the recessive gene for virulence (pathogenicity) in the pathogen

R= represents the dominant gene for resistance

‘r’ = the recessive gene for susceptibility in the attacked plant.

Host	Pathogen			
	a_1a_2	a_1A_2	A_1a_2	A_1A_2
r_1r_2	+	+	+	+
R_1r_2	+	+	—	—
r_1R_2	+	—	+	—
R_1R_2	+	—	—	—

Key: ‘—’ incompatibility (no disease development)

‘+’ compatibility (disease development)

6.5. Genetics of Resistance in Hosts

Plants are resistant to certain pathogens because they belong to taxonomic groups that are outside the host range of these pathogens (non host resistance), because they possess genes for resistance (R genes) directed against the avirulence genes of the pathogen (true, race specific, cultivar-specific, or gene-for-gene resistance), or because, for various reasons, the plants escape or tolerate infection by these pathogens (apparent resistance).

I. Apparent Resistance

A. Tolerance

Tolerance to disease is the ability of plants to produce a good crop even when they are infected with a pathogen. Tolerant plants are, obviously, susceptible to the pathogen, but they are not killed by it and generally show little damage.

B. Avoidance (Disease Escape)

Disease escape occurs whenever genetically susceptible plants do not become infected because the three factors necessary for disease (susceptible host, virulent pathogen, and favorable environment) do not coincide and interact at the proper time or for sufficient duration.

Eg. LBP----occurs on potato plants during cool and humid environment so growing potato during dry season with irrigation results in no disease development.

II. True Resistance

Disease resistance that is controlled genetically by the presence of one, a few, or many genes for resistance in the plant is known as true resistance. The host and the pathogen are incompatible because:-

- Lack of chemical recognition between the host and the pathogen
- The host plant can defend itself against the pathogen by various defense mechanisms already present or activated in response to infection by the pathogen.

There are two kinds of true resistance:

A. Horizontal Resistance

- Sometimes called partial, race nonspecific, general, quantitative, polygenic, adult-plant, field, or durable resistance.
- Controlled by many genes, thereby the name polygenic or multigene resistance.
- Effective against all races of pathogens
- Not affected by environments
- The resistance has range of possible expression from very low to very high level of resistance.
- Low selection pressure on pathogen

B. Vertical Resistance

- It is resistance to one or few races of pathogen
- It affected by environments (temporary)

- Controlled by one or few genes
- Are races resistance
- Are qualitative resistance
- The resistance has no range rather expressed either present or absent (no moderate resistance)
- High selection pressure on pathogen

6.6. Breakdown of Plant Disease Resistance

Varieties with specific (monogenic or oligogenic) resistance shows complete resistance to a specific pathogen under most environmental conditions, but a single or a few mutations in the pathogen may produce a new race that may infect the previously resistant variety.

★The breakdown in vertical resistance can be avoided by use of multilines. Multilines are either mixtures of individual varieties (lines or cultivars) that are agronomically similar but differ in their resistance genes or varieties that contain the mixtures of genes. E.g. small cereals against rust fungi.

★Horizontal resistance is universally present in wild and domesticated plants but it is at its highest levels in wild plants and is at its lowest in greatly 'improved varieties'.

⇒HR is affected by different environmental conditions, but a pathogen will undergo many more mutations to completely breakdown the resistance of the host.

⇒HR is eroded in the absence of the pathogen because there is no selection pressure for resistance.

⇒HR can be incorporated in to cultivated crops by crossbreeding of existing, genetically different varieties.

Note: As a rule, a combination of major and minor genes for resistance against a pathogen is the most desirable make up of any plant variety.

Chapter 7

Epidemiology of affecting plant disease

Epidemiology is the study of properties of pathogens, hosts, and the environment that lead to an increase in disease in a population.

Plant disease epidemics develop as a result of the interaction of susceptible host plants, a virulent pathogen, and favorable environmental conditions over a fairly long period of time. Although the four factors are involved in disease development, there are situations of the factors where by disease are developed.

7.1. Elements of Epidemics

Plant disease epidemics develop as a result of the timely combination of the same elements that result in plant disease: susceptible host plants, a virulent pathogen, and favorable environmental conditions over a relatively long period of time. Although the four factors are involved in disease development there are situations of the factors whereby diseases are developed:

Host plant factors

- ✓ Susceptible host
- ✓ Susceptible stage
- ✓ Predisposition

- **Pathogen factors**

- ✓ Virulent pathogen
- ✓ Infective stage
- ✓ Inoculum potential
- ✓ Age of inoculum

- **Environmental factors**

- ✓ Favorable environment to the pathogen
- ✓ Duration of the exposure
- ✓ Season of the year
- ✓ Soil status
- ✓ Ecology

- **Human factors**

- ✓ Cultivation practices
- ✓ Sources of seeds
- ✓ Introduction of susceptible host

When these conditions are satisfied the disease can thus:-

- Wipe out the plant eg. Chestnut Blight
- Cause famine eg. Irish potato famine
- Have huge economic failure eg. Coffee leaf rust

7.2. MEASUREMENT OF PLANT DISEASE AND OF YIELD LOSS

When measuring disease, one is interested in measuring:-

- **Incidence of the disease**, i.e., the number or proportion of plant units that are diseased (i.e., the number or proportion of plants, leaves, stems, and fruit that show any symptoms) in relation to the total number of units examined;

$$DI = \frac{\text{Number of infected plants}}{\text{Total number of plants}} \times 100$$

- **Severity of the disease**, i.e., the proportion of area or amount of plant tissue that is diseased;
 - ✓ Usually expressed as the percentage or proportion of plant area or fruit volume destroyed by a pathogen. More often, disease assessment scales from 0 to 10 or 1 to 4 are used to express the relative proportions of affected tissue at a particular point in time.

- **Yield loss caused by the disease**, i.e., the proportion of the yield that the grower will not be able to harvest because the disease destroyed it directly or prevented the plants from producing it (the yield loss is the difference between **attainable yield** and **actual yield**).

Yield loss due to disease is measured at a specific growth stage, from sequential disease assessments at several stages of a crop's growth, or by determining the area under a disease progress curve (AUDPC).

AUDPC: - the area between the disease progress curve and the X axis of the graph. AUDPC is used to summarize the progress of disease severity and is calculated by a formula that takes into account the number of times the disease severity was evaluated, the disease severity at each evaluation time, and the time duration of the epidemic.

$$\text{AUDPC} = \sum_{i=1}^{n-1} \left[(y_i + y_{i+1}) / 2 \right] (t_{i+1} - t_i)$$

Eg. Calculate the AUDPC the following data

Severity (Y)		Time (t)	
1	2		
3	2	10 AE	AUDPC₁= 230
5	4	20	AUDPC₂=240
9	10	30	which one is more resistance?
15	18	40	

- ❖ **For cultivar when AUDPC is high; indicates low resistance but for pathogens when AUDPC is higher; indicates its high virulence**

Yield loss almost always results in economic loss from disease. Economic loss occurs whenever economic returns from the crop decrease because of reduced yields

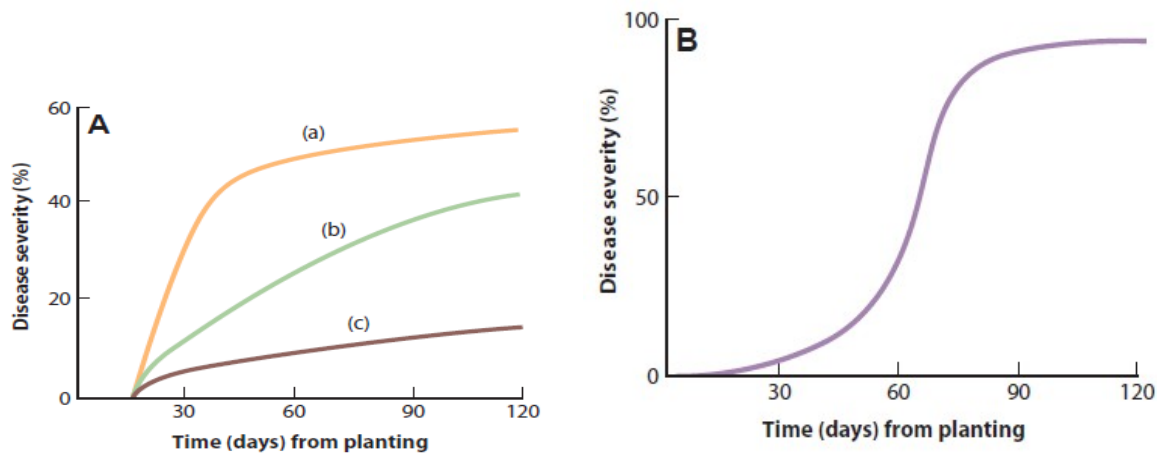
The level of disease, i.e., the amount of plant damage, at which control costs just equal incremental crop returns is called **the economic threshold of the disease**.

7.3. Disease Gradient Curve

A curve which shows the pattern of epidemic interns of changes in the number of lesions, amount of diseased tissue and the number of plants.

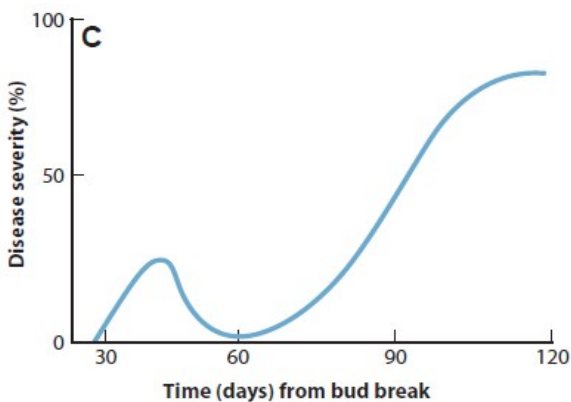
The point of origin and the shape of a disease–progress curve reveal information about the time of appearance and amount of inoculum, changes in host susceptibility during the growing period, recurrent weather events, and the effectiveness of cultural and control measures.

Knowledge of disease–progress curves also allows disease forecasting, the most susceptible stage of the plant and selection of the best control strategy for the particular disease and time.



(A) Three monocyclic diseases of different epidemic rates

(B) Polycyclic disease



(C) Bimodal polycyclic disease

Comparison of epidemics

What is the importance and how to compare diseases epidemics?

It is important to determine the relative resistance of different plant species and to determine the relative aggressiveness of the pathogen.

How?

To determine its importance

- ❖ Inoculation
 - ✓ A given pathogen to different plant cultivars
 - ✓ Different race of pathogen to different cultivars
- ❖ Disease assessment

7.4. Epidemic rate of plant diseases

i. Monocyclic diseases

- ✓ 1⁰ inoculum not lead to 2⁰ infection
- ✓ The increase in disease is mainly due to 1⁰ inoculum
- ✓ Infection does not result in another infection and they are called simple interest diseases.
- ✓ Latent period (time gap b/n inoculation to sporulation) is long
- ✓ Eg. Smut, root rot vascular wilt, tree rust

ii. Polycyclic diseases

- ✓ Disease which have many cycle per year (20-30 cycle/year)
- ✓ Compound interest diseases
- ✓ Infection will lead to another infection
- ✓ Highly problematic disease for plant producer
- ✓ Latent period is short
- ✓ Eg. LBP, powdery mildew, leaf spot, light, insect born virus

7.5. Monitoring and Forecasting Plant Disease Epidemics and Bases for Forecasting

Forecast: estimation in advance or prediction of a future event or condition. The forecasting of epidemics is, therefore, simply a prognostication or foretelling. Forecasting involves all the activity in ascertaining and notifying the growers of a community that conditions are sufficiently favorable for certain diseases.

Disease forecasting allows the prediction of probable outbreaks or increases in intensity of disease and, therefore, allows us to determine whether, when, and where a particular management practice should be applied. In managing the diseases of their crops, growers must always weigh the risks, costs, and benefits of each of the numerous decisions. Most frequently, farmers need forecasts:

- ✓ **to determine whether a plant infection is likely to occur or not**
- ✓ **to decide the amount of chemicals and labor used without increasing the risk of losing their crop.**
- ❖ To develop a plant disease forecasting system, the particular pathogen, host plant, and environment must be considered.

Bases for forecasting Plant Diseases

1. Amount of initial inoculum
2. Amount of secondary inoculums
3. Weather conditions favoring and secondary inoculum or vectors
4. The host plant (degree of susceptibility of varieties).

Chapter 8

GENERAL CONTROL METHODS OF PLANT DISEASES (5)

The word ‘control’ is a complete term where permanent ‘control’ of a disease is rarely achieved whereas, ‘management’ of a disease is a continuous process and is more practical in influencing adverse affect caused by a disease. Disease management requires a detail understanding of all aspects of crop production, economics, environmental, cultural, genetics and epidemiological information upon which the management decisions are made.

8.1. Regulatory Disease Control Methods

National and state laws regulate the import and distribution of crops between states and countries.

Such regulatory control methods include

1. Quarantine
2. Inspections of plants in the field/warehouses
3. Compulsory eradication of host plants
4. Exclusion (elevation)

8.1.1. Quarantines and Inspections

Quarantine refers to the control of import and export of plants to prevent (Prophylactic measures) spread of diseases and pests. This method excludes the pathogen from entering in to the area or country. Plant quarantine measures can keep out foreign pathogens from entering in to the area or entry in to or passage through the second country, i.e., importing country, via plants, plant products, soil, and other materials. Quarantine regulations are enforced in most countries, e.g. at the entry port to intercept catastrophic diseases. Quarantine is effective against viruses, bacteria, nematodes and parasitic higher plants carried through seeds and plant materials. Plant quarantine measures include:-

1. Growing plants under observations for certain periods before releasing to the importer.
2. Inspecting the crop fields for the occurrence of potentially dangerous pathogens. If they are free from diseases, inspection certifies are issued.
3. Controlling the movement and sale of seeds between states (However, they are not effective against air-born pathogens).
4. Indexing of crops in the field and storage. Indexing is a procedure to determine whether a virus can infect a given plant. It involves the transfer of bud, scion, sap, etc. from one plant to one or more kinds of (indicator) plants that are sensitive to the virus.

Certification of Plant Materials: - Certification is normally offered on the basis of purity, freedom from parasitic weeds, and germination percentage for which standards are fixed for each crop. Seeds are treated with pesticides. For vegetative propagative stocks, tubers, and other planting materials, plants are examined in the growing season for freedom from diseases. Heat

treatment, meristem, or tuber indexing are practiced to make such planting materials free from of pathogens.

A. Principles of plant disease management

- i. **Avoidance of the pathogen:** Occurrence of a disease can be avoided by planting/sowing a crop at times when, or in areas where, inoculums remain ineffective /inactive due to environmental conditions, or is rare or absent.
- ii. **Exclusion of the pathogen:** This can be achieved by preventing the inoculums from entering or establishing in a field or area when it does not exist. Legislative measures like quarantine regulations are needed to be strictly applied to prevent spread of a disease.
- iii. **Eradication of the pathogen:** It includes reducing, inactivating, eliminating or destroying inoculums at the source, either from a region or from an individual plant (rouging) in which it is already established.
- iv. **Protection of the host:** Host plants can be protected by creating a toxin barrier on the host surface by the application of chemicals.
- v. **Disease resistance:** Preventing infection or reducing the effect of infection of the pathogen through the use of resistance host which is developed by genetic manipulation or by chemotherapy.
- vi. **Therapy:** Reducing severity of a disease in an infected individual.

The first five principles are prophylactic (preventive) procedure and the last one is curative.

B. Methods of plant disease management

1. Avoidance of the pathogen:

- i. Choice of geographical area
- ii. Selection of a field
- iii. Adjustment of time of sowing
- iv. Use of disease escaping varieties
- v. Use of pathogen-free seed and planting material
- vi. Modification of cultural practices

2. Exclusion of inoculums of the pathogen

- i. Treatment of seed and plating materials
- ii. Inspection and certification
- iii. Quarantine regulations
- iv. Eradication of insect vector

3. Eradication of the pathogen

- i. Biological control of plant pathogens
- ii. Eradication of alternate and collateral hosts
- iii. Cultural methods:
 - a. Crop rotation
 - b. Sanitation of field by destroying/burning crop debris
 - c. Removal and destruction of diseased plants or plant parts
 - d. Rouging
- iv. Heat and chemical treatment of diseased plants
- iv. Soil treatment: by use of chemicals, heat energy, flooding and fallowing.

4. Protection of the host

- i. Chemical control: application of chemicals (fungicides, antibiotics) by seed treatment, dusting and spraying
- ii. Chemical control of insect vectors
- iii. Modifications of environment
- iv. Modification of host nutrition

5. **Disease resistance:** Use of resistant varieties: Development of resistance in host is done by:-

- i. Selection and hybridization for disease resistance
- ii. Chemotherapy
- iii. Host nutrition
- iv. Genetic engineering, tissue culture

6. **Therapy:** - Therapy of diseased plants can be done by

- i. Chemotherapy

- ii. Heat therapy
- iii. Tree-surgery

8.2. CULTURAL METHODS

Modifying the environment to protect plants is usually accomplished by altering cultural practices. Farmers alter practices expecting diseases losses will be reduced but improper, excessive, or untimely usage might increase losses. Cultural practices used to control or manage plant diseases include application, manipulation, or modification of any practice involved in crop production.

These methods encompass:

i. **Phytosanitation**

- Planting disease free planting materials which is important to control the Carry over population
- Clearing fields from unwanted plants-important especially for obligate parasite like rust pathogens
- Disinfection of agricultural tools

ii. **Choose of variety**

- Select varieties which have horizontal vs vertical resistance
- Easy to apply, ecologically free, relatively sustainable and cheaper

iii. **Choose of planting date**

- **No dieses** unless the susceptible stage of the crop matchs to the virulent stage of the pathogen.
- **Eg. From July to April** hence the condition is cool and wet it is difficult for potato production due to LBP but from **May to early June** production is safe.

iv. **Manipulating the environments**

- Especially the micro climate eg. Disease of Root Rot caused by Pythium needs water logged condition so by drain the area it is possible to control.

v. **Crop rotation**

Is an especially effective **cultural control** method that can reduce pathogen populations and increase crop yields. But rotation is only useful against pathogens that produce inoculum that is not long-lived.

Other cultural practices also can be used to control diseases. These include **deep plowing** to bury potential sources of pathogen inoculum, and manipulation of soil physical and chemical properties to make the field environment less conducive to survival of inoculum and disease development.

8.3. Biological Control of Plant diseases

Refers the use of living organisms (bioagent) which is important to reduce the pathogen below the economic thresh hold levels.

- ❖ As disadvantage they may change their behavior, needs high initial investments and slower action
- ❖ As advantage the method is safe, self perpetuating, sustainable and cheap after initial investment

Steps in using biological agents for disease control

- Identification of the problem
- Searching for potential bioagent
- Selection (bioagent having narrow host range, adaptive for the prevalent ecosystem, optimum reproductive capacity)
- Introduction
- Rearing
- Releasing
- Monitoring and evaluation

8.4. Chemical Controls

This method deals with use of chemical compounds that are toxic to the pathogens. It is the most commonly used method of controlling plant diseases in the field, in the storage and in the greenhouse. The chemical compound may:

1. Inhibit spore/mycelium germination in fungi or interfere with egg hatching in nematodes;

2. Inhibit the growth of pathogens;
3. Inhibit multiplication or sporulation of pathogens; and
4. Be lethal to the pathogens due to their toxicity.

Based on the pathogens controlled, the chemicals could be fungicides, bactericides, nematocides, “viricides”, or herbicides for parasitic higher plants. Most chemicals are used to disinfect the soil, warehouses, treat tree wounds, or protect stored fruits, vegetables and grains, or control pathogen vectors. Chemicals could be contact (local action) or systemic fungicides and antibiotics (absorbed and translocated) through the plant system.

8.4.1. Uses of Chemical Application

1. Foliar treatment and Dusts

Fungicides and bactericides are applied in these ways. Most fungicides and bactericides are protectants and must be present on the surface of the plants in advance of the pathogen to prevent infection. The chemicals inhibit spore germination or kill bacteria and inhibit multiplication.

Dusting is rarely used nowadays as it is much less efficient than spraying, especially in terms of persistence. However, sulfur dust in particular is occasionally still used on some crops such as grapes. Although a major advantage of dusting is that obviates the need to transport water, the relative cost of active ingredient is very much higher due to the cost transporting a high proportion of inert filler.

Applying by dust or spray before or immediately every rain. The whole plant part must be covered to obtain satisfactory results from contact fungicides/bactericides (protectants), especially, young, expanding leaves, twigs, and fruits. Interval of sprays varies from 7-14 days or longer. This depends on (1) particular disease (2) frequency and duration of rain (3) persistence (residue life) of fungicide (4) season of year. Optimal number of sprays per season varies from 2 to 15 or more. Dusters, sprayers and fumigators are used for chemical application on seed, foliage and soil.

2. Seed treatment

Seed treatment is the use and application of biological and chemical agents that control or contain primary soil and seed borne infestation of insects and diseases which pose devastating

consequences to crop production and improving crop safety leading to good establishment of healthy and vigorous plants resulting better yields.

3. Soil treatment

Certain fungicides are applied as fumigants (volatile compounds), dusts drenches, or granules to control nematodes, fungi and bacteria by reducing the inoculum. Thus damping-off, seedling blights, wilts, crown and root rots, and other diseases are controlled. These fungicides include captan, diazoben, PCNB, chloroneb, metalaxyl, fosetyl-Al, triadimefon, ethazol, and propamocarb. Many of the systemic fungicides such as metalaxyl, fosetyl-Al, are so effective even in small amounts that a single preplant broadcast soil spray followed by disking can provide a full season's protection against several pathogens. Fumigation is also possible to control soil born diseases.

4. Treatment of Tree Wounds

Painting tree wounds with shellac or with any other commercial wound dressing prevents the trees from infection by pathogens. The wood cuts are also sterilized by swabbing them with a solution of 0.5- 1.0% NaCl (10-20% Clorox) or 70% ethyl alcohol. Tree wounds are also painted with a 10:2:2 lanolin, rosin, and gum mixture (or Cerano), Bordeaux paint, or with asphalt varnish tree paint.

8.4.2. Chemical Formulations

Formulation determines how a pesticide is used. The following formulations are used in pesticide application.

1. Powders/Dusts (D)

The particle size of the dusts range between 15-25µm for surface treatment and 25-50µm (even 70 µm) for aerial treatment, but poorly retained in the soil. They are mechanical mixtures of fungicide and filler or carrier. They are used at a rate of 10-30 kg/ha dust. Talc, chalk, kaolin, diatomite, silica gel, pyrophyllite and clays are used as carriers. From 3-5% mineral oil may be added to dusts to reduce drift.

2. Wettable Powders (WP)

They contain 30-80% active ingredients (A. I.), 15 to 60 % carriers and 2.5.to 4% others such as adhesives and surfactants. About 80% of the particles are 3 μm in diameter. When diluted with water, they give stable suspensions and better adhere on foliage than dusts. Good WPs must fulfill the following characteristics.

- A. They must be stable in storage and not cake;
- B. They must rapidly form suspensions with slow setting of the solid particles; and
- C. They must wet the leaves of plants very well and be retained.

3. Granules (G)

The diameters of the particles range between 0.25 and 5 mm, the large ones with 3-5 mm and small ones with 0.25 -1.5 mm. Granules are incorporated in to the soil to control soil fungi and other pathogens. Granules contain toxicants, carriers, and adhesives. Nematicides are generally formulated as granules. Granules generally lose their pesticidal action less than other formulations; can pose less danger to man, and cause less phytotoxicity.

4. Solutions (S)

These formulations are dispersed in water and in organic solvents. They contain surfactants to reduce surface tension to wet the leaf surface well. Petroleum hydrocarbons, deodorized kerosene, diesel fuel, or mineral oils are used to prepare these pesticide solutions.

5. Emulsifiable Concentrates (EC)

These formulations are solutions of pesticide in oil comminuted in to fine drops coated with an emulsion that do not stratify for a long time. Homogenizers are used to prepare EC. Emulsifiable concentrates are divides into miscible and immiscible concentrates. The miscible ones contain fungicide, solvent, and emulsifier. They are prepared at 40-80 $^{\circ}\text{C}$ temperature.

6. Aerosol (A)

The droplet size varies from 0.001 to 50 μm (optimum 20-50 μm) in diameter. Aerosols are defined as suspensions of fine droplets of liquid or solid particles of pesticides in air.

7. Fumigants (F)

These are gases used in the decontamination of objects from insects, their larvae, and eggs, mites, nematodes, and other pests by fumigating them with vapour or gases of poisonous substances.

8.4.3. Methods of Application and Mechanisms of Action

1. Spraying

This method is a widely used way of applying pesticides in the form of solutions, emulsions, or suspensions to a surface being treated. Special equipment called sprayer (hand, tractor, and aerial) is used for spraying pesticides. Spraying is a universal method of pesticide application. Advantages of spraying:

1. It uses very low active ingredients per unit area;
2. It has uniform distribution and good coverage of treated surfaces; and
3. There is very good pesticide retention on treated surfaces, especially if wetting agents and adhesive are contained

2. Dusting

Dusting is the application of dust formulations to the surface of plants. Dusting can cover dense cereal crops very well. Disadvantages of dusting include:

1. High dust contact of the air is dangerous to workers
2. It requires high rate of the formulation
3. There is drift of dust cloud with wind currents, especially in those having 25µm diameter of particles
4. There could be a problem of washing off by rain.

3. Fumigation

Fumigation is one of the most widespread ways of controlling pests in storage and transportation, pests and diseases in sheltered ground, planting stock and valuable citrus crops and tea. Fumigation is very effective method because the toxicant vapour (gases) together with the air penetrates very well in to various porous materials, cracks, and minute openings.

8.4.4. Advantages and Disadvantages of Chemical Control

A. Advantages of Chemical Control

1. Fungicides can inhibit germination, growth, and multiplication of pathogens as contact or systemic chemicals.
2. Chemicals show very quick results in controlling plant diseases (protectants, eradicants, and fumigants).
3. Chemicals are used to disinfect stores, soils, and are as protectants of planting materials such as seeds.
4. Fungicides can be employed in prophylactic (preventive) programmes.

B. Disadvantages of Chemical Control

1. Pesticides are, in general, expensive; their application is also labor and time consuming.
2. Pesticides require trained manpower for application and costs for pesticide applicators are required.
3. Development of pesticides requires a long time and high resources.
4. Most chemicals used in crop protection are toxic to human beings and animals.
5. Some chemicals are not convenient for storage, i.e. they could be flammable, corrosive to metal containers, volatile, or may have short shelf life.
6. Some chemicals have phytotoxicity problems to sensitive plants
7. Some pathogens and pests can develop resistance to the chemicals and make them useless for future use.

Strategies to avoid the development of fungicide resistance

❖ Reduce fungicide use:

- Apply fungicides only when and where necessary, ideally based on a disease risk prediction model.
- Use fungicides as part of an integrated control programme, e.g. combined with disease-resistant cultivars and cultural measures to reduce inoculum.

❖ Diversify fungicide treatments:

- Avoid repeated use of fungicides with the same mode of action.

- Use mixtures of fungicides with different modes of action.
- In a spray program, alternate fungicides with different modes of action.
- Include multisite fungicides in mixtures or alternations.

8.5. INTEGRATED CONTROL OF PLANT DISEASES

The goal is to attain a sustainable system for control and management of disease that is environmentally friendly and economically sound. The plant pathologist needs to put together a package of control measures that can be presented to the farmer, including the use of resistant varieties, choice of best planting dates, crop rotations, sanitation methods, cultural methods, biological control and chemical controls.

Ideally, the integrated management strategy will be designed to accommodate several diseases simultaneously. The goal is to optimize crop yield and economic benefits while minimizing environmental impacts, leading to a sustainable agricultural ecosystem.

Disease control programs are based on an understanding of host and pathogen biology and the factors involved in infection and disease development. Short-term control may be achieved by a single practice. However, long-term reduction of disease losses requires the implementation of an integrated control program, including the use of disease-resistant hybrids, crop rotation, tillage, balanced fertility, insect and weed control, and, if necessary, the timely application of disease-control chemicals.

Disease-control practices should be based on knowledge of which diseases are likely to occur and when they are likely to cause economic loss. Field surveys can be helpful in identifying diseases and developing appropriate disease-control measures. For example, fall surveys for stalk and ear rots will help reduce losses from these diseases and determine drying and storage strategies.

Generally, the goal of plant disease control includes:

1. Elimination or reduction of the initial inoculum
2. Reduction of the effectiveness of initial inoculum
3. Increase in the resistance of the host

4. Delaying the onset of disease
5. Slowing down of the secondary of secondary rules

REFERENCES

- Agrios, G.N. 1997 & 2005. Plant pathology, 4th & 5th editions. Academic Press: New York. 635 pp & 922 pp. (Text book).
- Alexopoulos, C. J., Mims, C.W. and Blackwell, M. 1996. Introductory mycology, 4th edition. John Willey & sons, Inc. New York.
- Chattopadhyay, S.B. 1980. Principles and procedures of plant protection, 2nd edition. Oxford and IBH publ.Co., New Delhi.
- Dickson, M. 2003. Molecular plant pathology. Bios Scientific Publishers: London. 244pp.
- Mehrotra, R.S. 2000. Plant pathology, 17th reprint. Tata MacGraw-Hill publ. Co. Ltd. New Delhi.
- Singh, R.S. 1999. Introduction to principles of plant pathology, 3rd edition. Oxford & IBH Publ. Co. Pvt. Ltd., New Delhi.